3DOD_snow visualization
3d-bbox on image

이상욱

요구사항 분석

Model_name : Scene_number

kitti_GT
(1)

Kitti_pred
(2)

Snow_pred
(3)

(1) Raw_kitti_GT : pointcloud of raw-kitti
(2) Kitti_pred : prediction(3d-bbox) of raw-kitti
(3) Snow_pred : prediction(3d-bbox) of snow-kitti

Bbox color info

	Car	Pedestrian	Cyclist
GT	초록색	하늘색(cyan)	노란색
Pred	빨간색	보라색(magenta)	주황색

눈

Pred o

Pred x

내용 구성

- 3d-bbox on point cloud
  – raw kitti (4-9p)
  – snow kitti (10-15p)

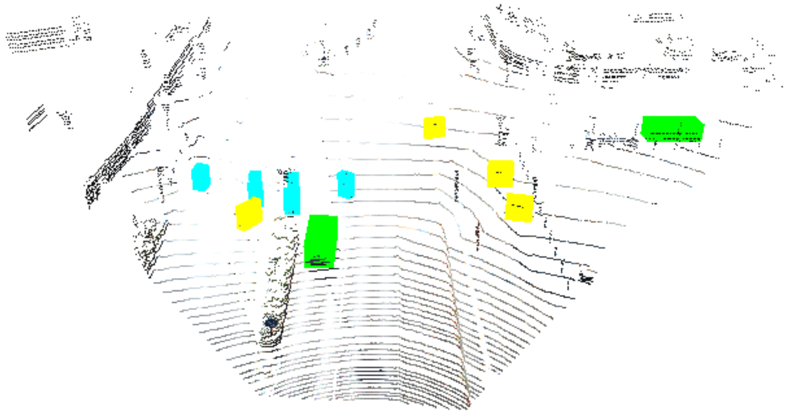
- Why doesn't snow simulation affect Pedestrian much?
  – evaluation results (17p)
  – bbox on images (19-29p)
  – point cloud removing ratio (31p)

3d-bbox on point cloud – raw kitti (4-9)

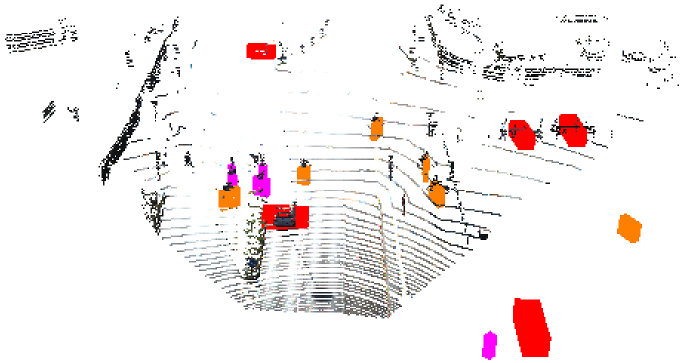
Scene_num : 006813

3d-bbox on point cloud – raw kitti

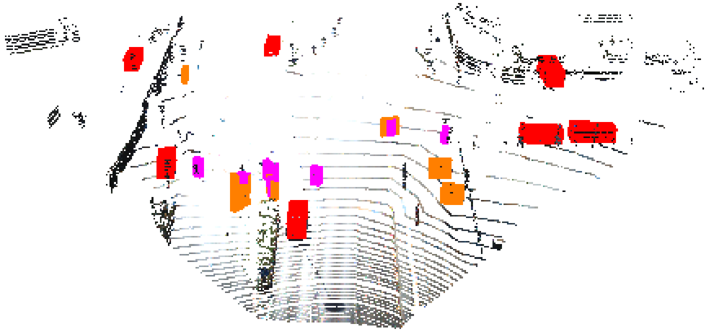
	Car	Pedestrian	Cyclist
GT	초록색	청록색(cyan)	노란색
Pred	빨간색	보라색(magenta)	주황색



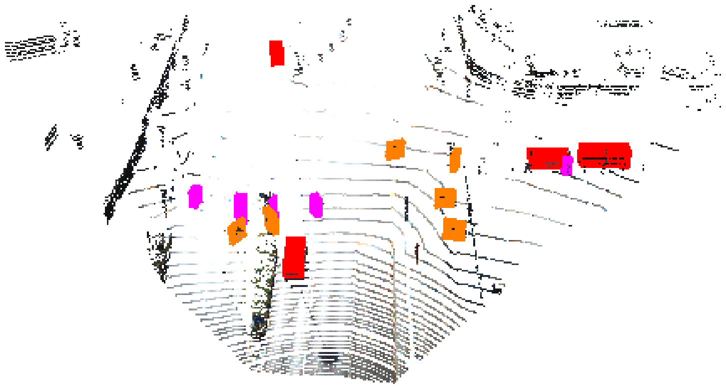
kitti_GT



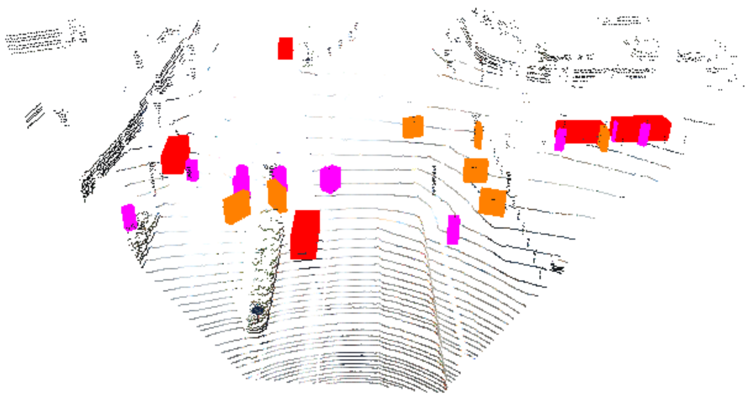
PointPillars



PointRCNN



VoxelRCNN

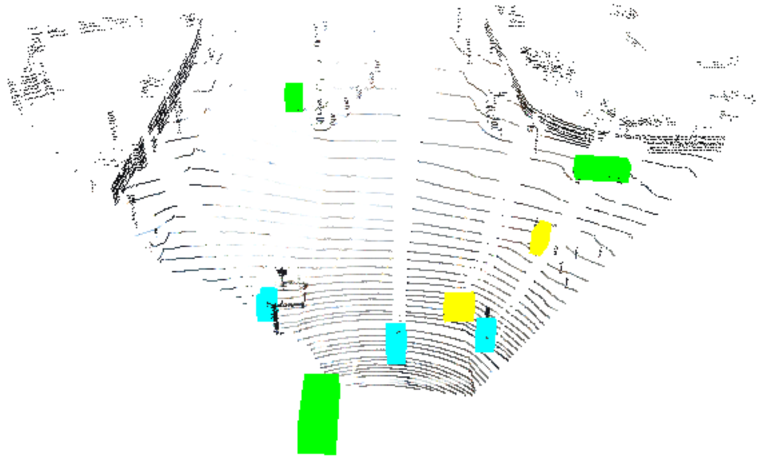


mvmm

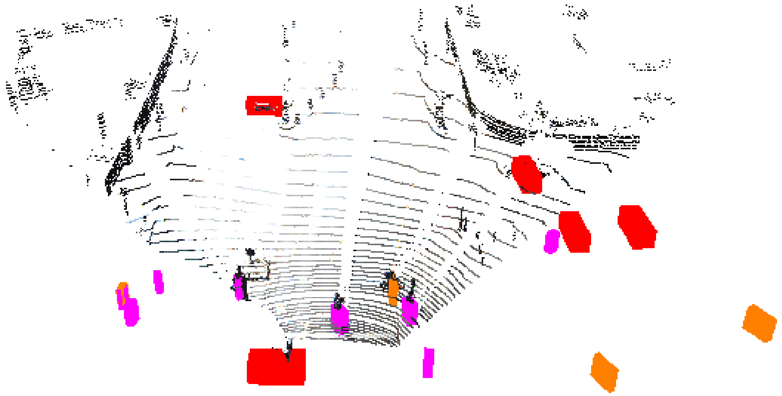
Scene_num : 006496

3d-bbox on point cloud – raw kitti

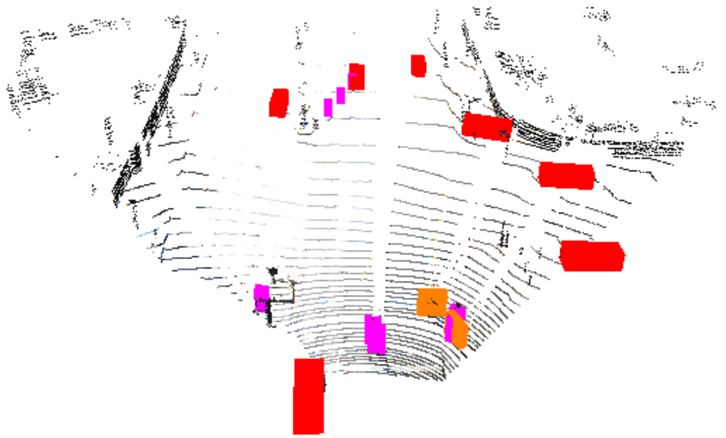
	Car	Pedestrian	Cyclist
GT	초록색	청록색(cyan)	노란색
Pred	빨간색	보라색(magenta)	주황색



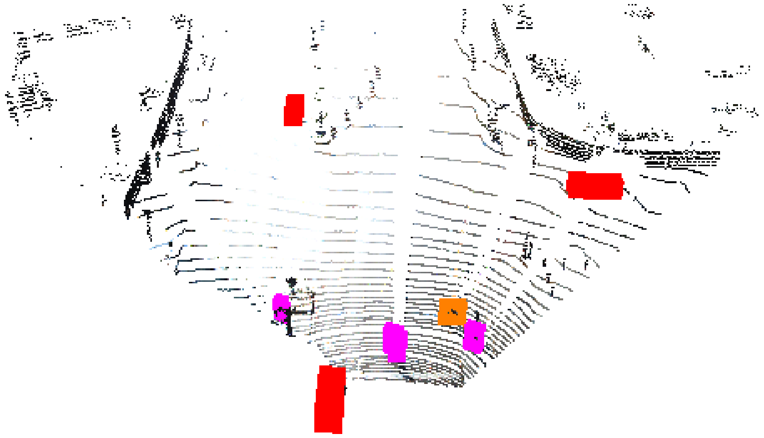
kitti_GT



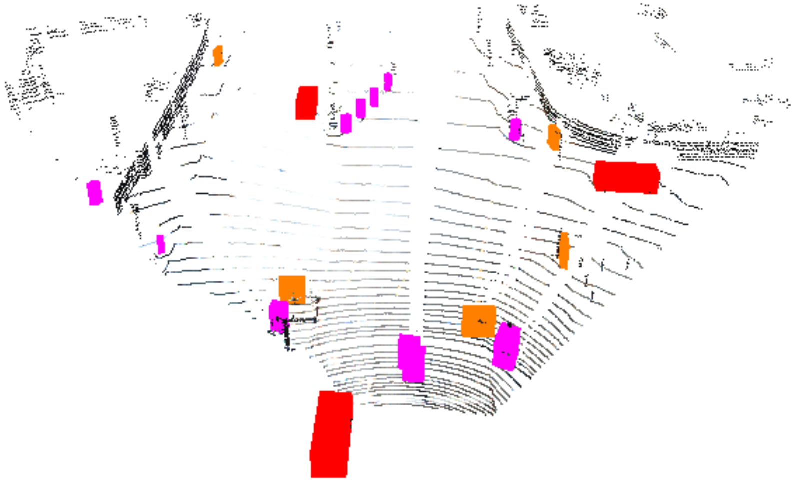
PointPillars



PointRCNN



VoxelRCNN

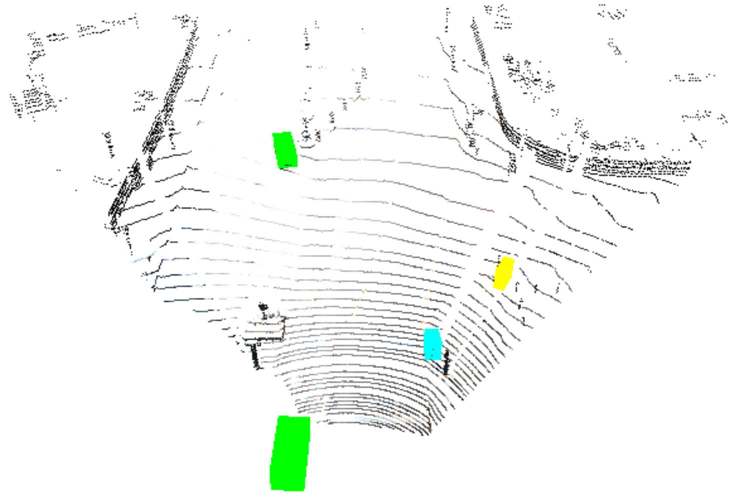


mvmm

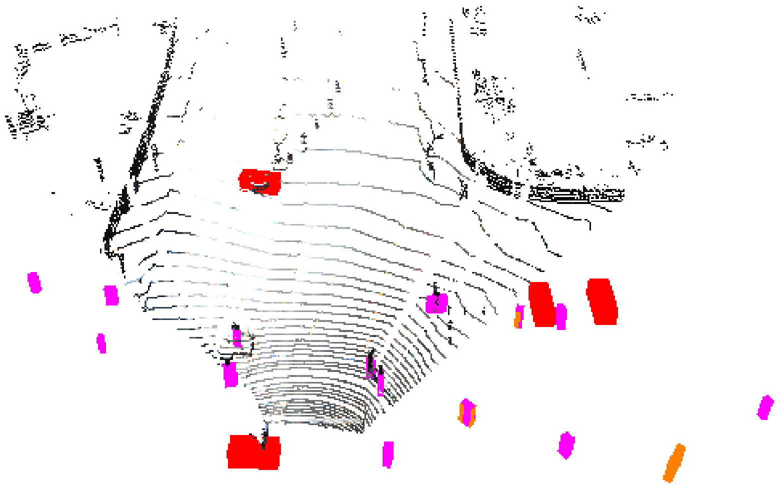
Scene_num : 006024

3d-bbox on point cloud – raw kitti

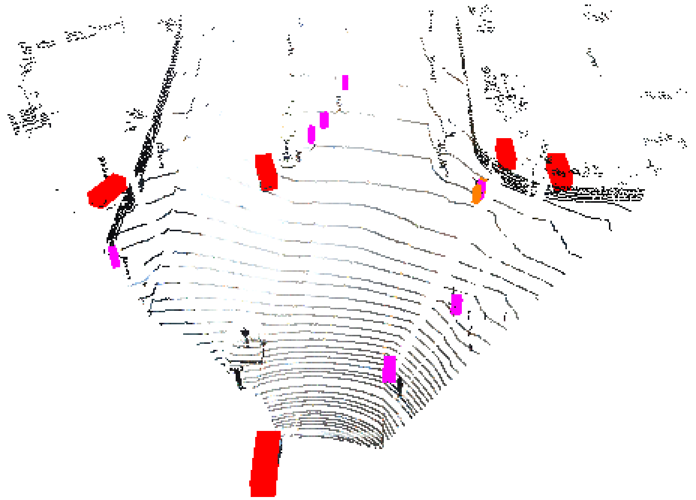
	Car	Pedestrian	Cyclist
GT	초록색	청록색(cyan)	노란색
Pred	빨간색	보라색(magenta)	주황색



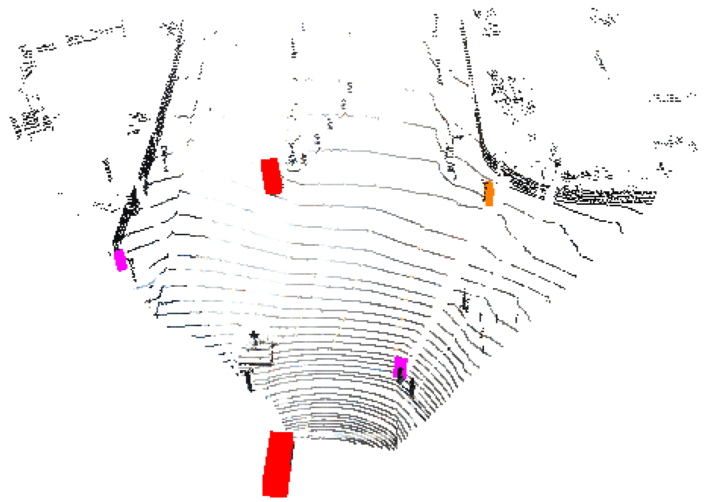
kitti_GT



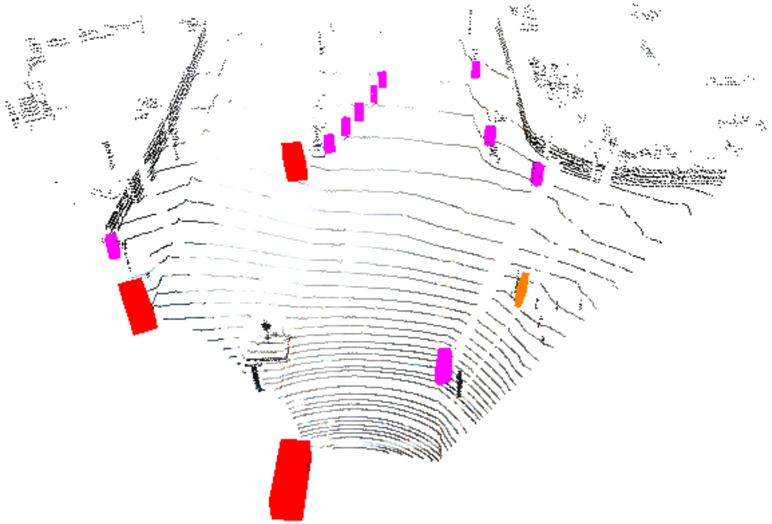
PointPillars



PointRCNN



VoxelRCNN



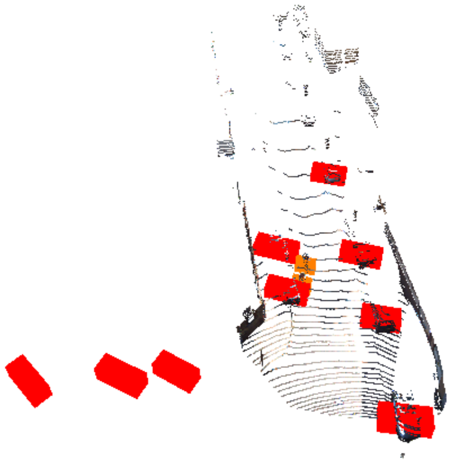
mvmm

Scene_num : 000201

3d-bbox on point cloud – raw kitti

	Car	Pedestrian	Cyclist
GT	초록색	청록색(cyan)	노란색
Pred	빨간색	보라색(magenta)	주황색

kitti_GT



PointPillars

PointRCNN

VoxelRCNN

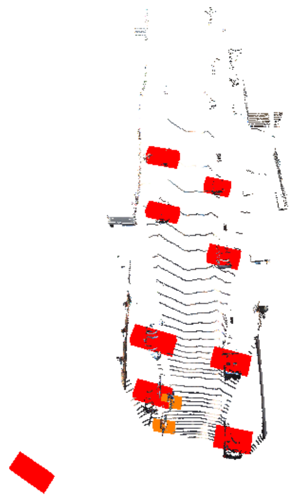
mvmm

Scene_num : 001221

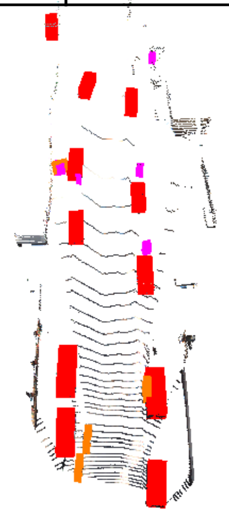
3d-bbox on point cloud – raw kitti

	Car	Pedestrian	Cyclist
GT	초록색	청록색(cyan)	노란색
Pred	빨간색	보라색(magenta)	주황색

kitti_GT



PointPillars



PointRCNN

VoxelRCNN

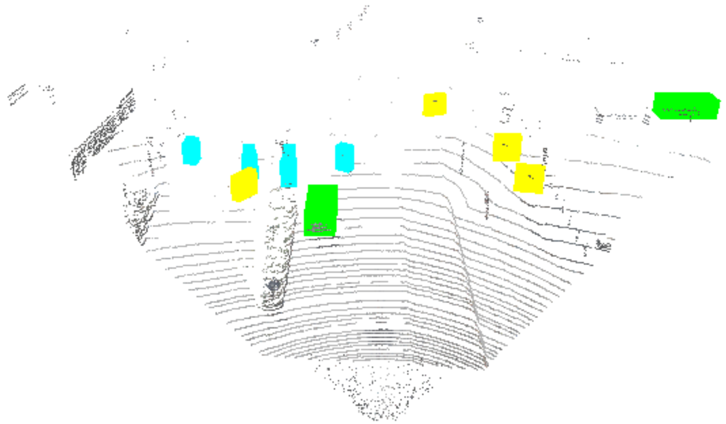
mvmm

3d-bbox on point cloud
– snow kitti (10-15)

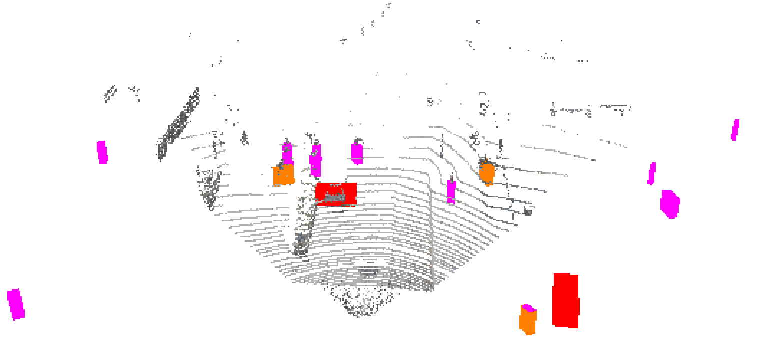
Scene_num : 006813

3d-bbox on point cloud – snow kitti

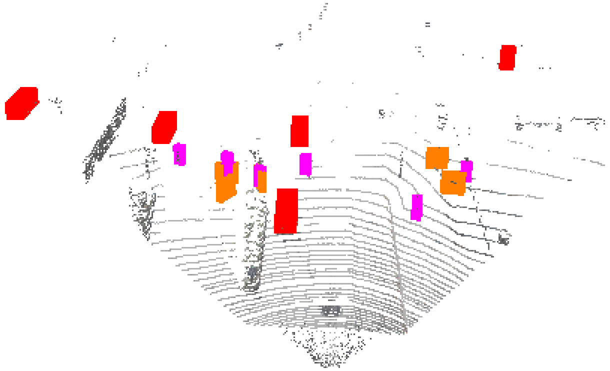
	Car	Pedestrian	Cyclist
GT	초록색	청록색(cyan)	노란색
Pred	빨간색	보라색(magenta)	주황색



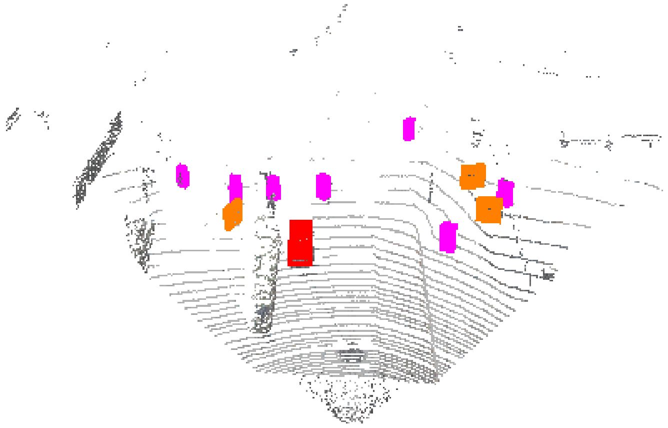
Snow_GT



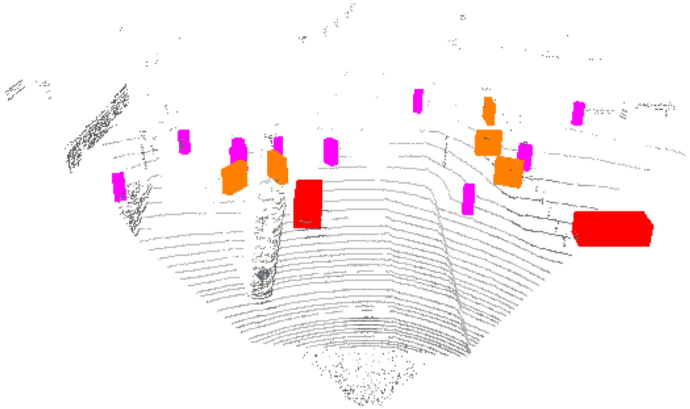
PointPillars



PointRCNN



VoxelRCNN

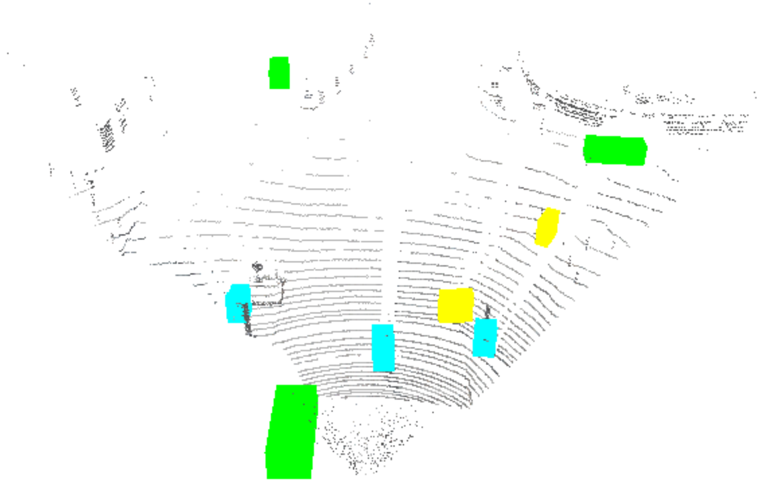


mvmm

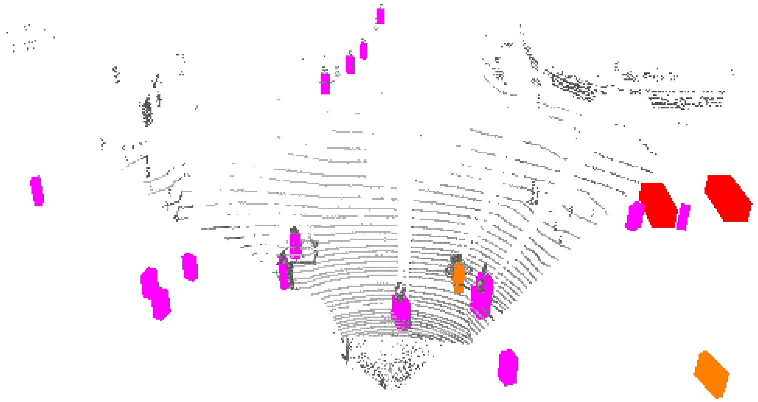
Scene_num : 006496

3d-bbox on point cloud – snow kitti

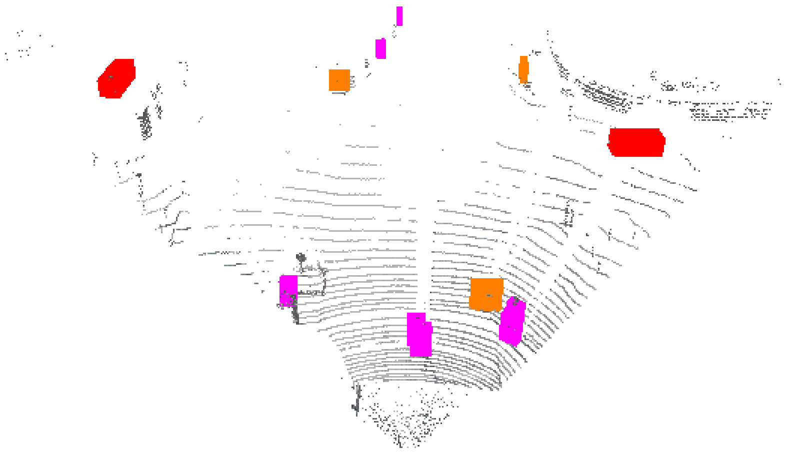
	Car	Pedestrian	Cyclist
GT	초록색	청록색(cyan)	노란색
Pred	빨간색	보라색(magenta)	주황색



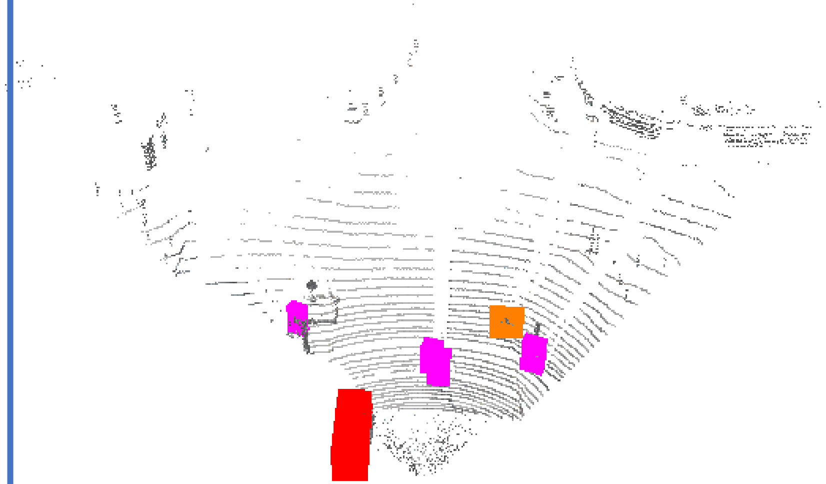
Snow_GT



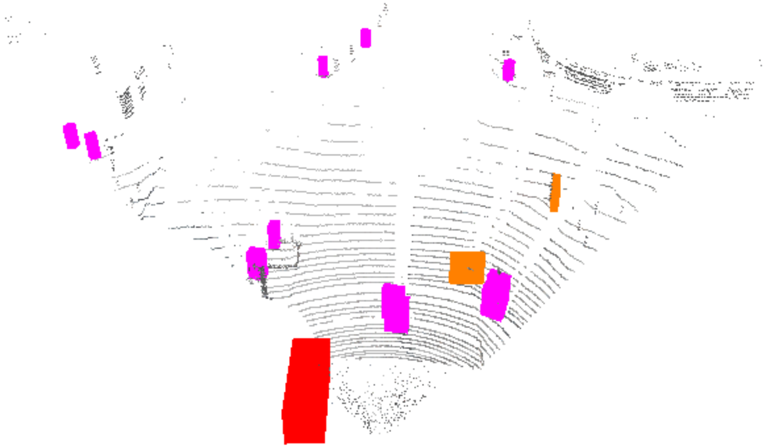
PointPillars



PointRCNN



VoxelRCNN

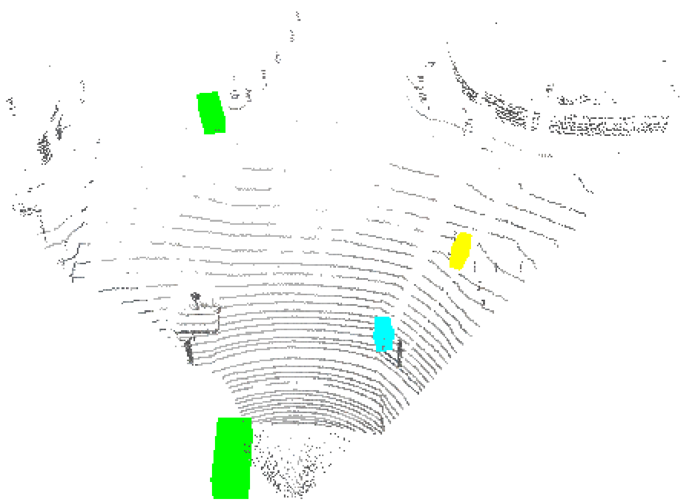


mvmm

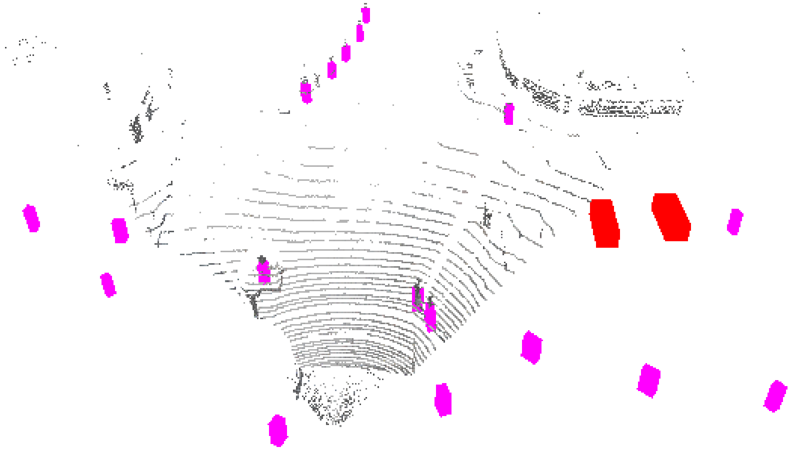
Scene_num : 006024

3d-bbox on point cloud – snow kitti

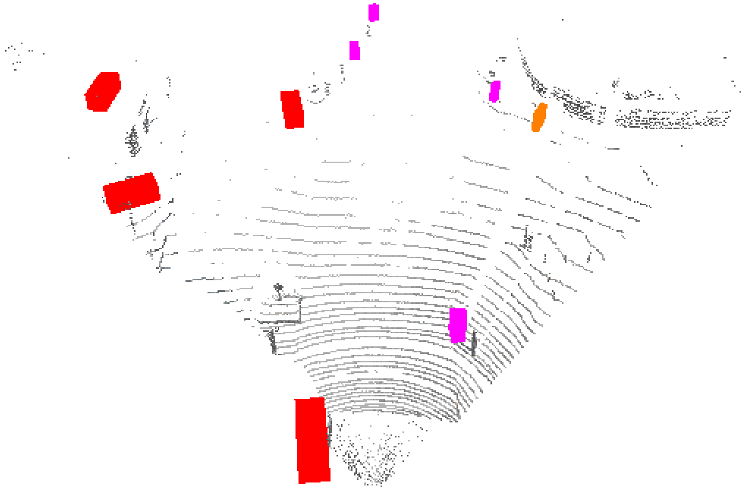
	Car	Pedestrian	Cyclist
GT	초록색	청록색(cyan)	노란색
Pred	빨간색	보라색(magenta)	주황색



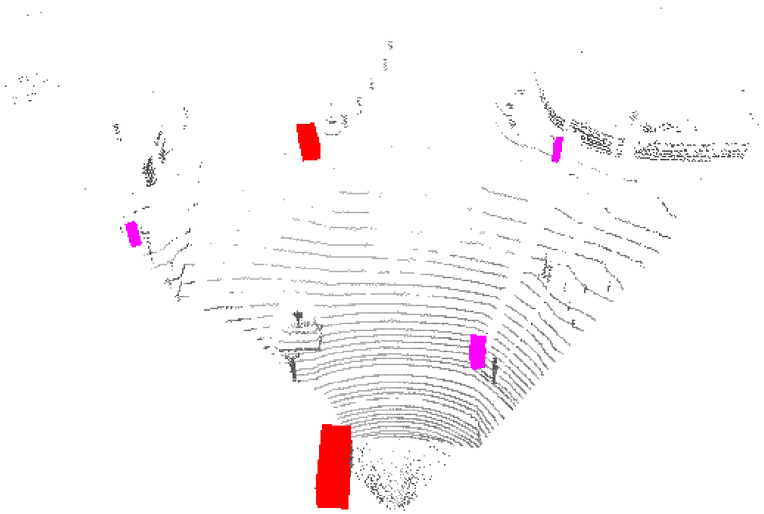
Snow_GT



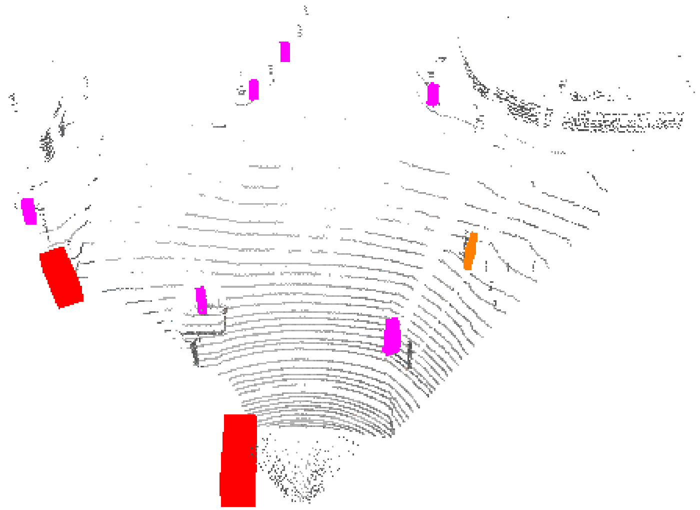
PointPillars



PointRCNN



VoxelRCNN



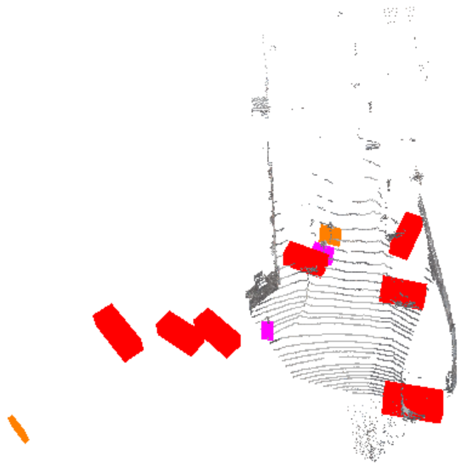
mvmm

Scene_num : 000201

3d-bbox on point cloud – snow kitti

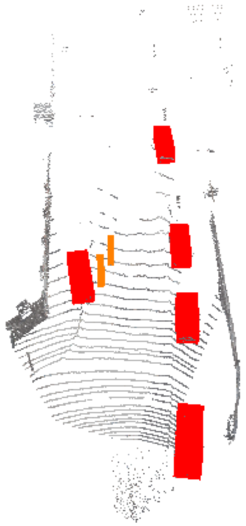
	Car	Pedestrian	Cyclist
GT	초록색	청록색(cyan)	노란색
Pred	빨간색	보라색(magenta)	주황색

Snow_GT



PointPillars

PointRCNN



VoxelRCNN

mvmm

Scene_num : 001221

3d-bbox on point cloud – snow kitti

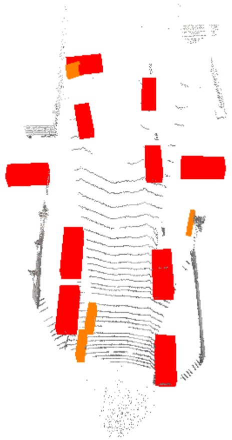
	Car	Pedestrian	Cyclist
GT	초록색	청록색(cyan)	노란색
Pred	빨간색	보라색(magenta)	주황색

Snow_GT

PointPillars

PointRCNN

VoxelRCNN



mvmm

Why doesn't snow simulation affect Pedestrian much?
- Evaluation result

Why doesn't snow simulation affect Pedestrian much? - evaluation

LiDAR-only

Repo	Metric	Pedestrian mAP	Cyclist mAP	Car mAP	GPU
PointPillars_kitti	3D-Bbox	51.4551 47.9575 43.8407	81.8821 63.6698 60.9022	86.6348 76.7492 74.1609	5060ti x1
PointPillars_kitti_snow	3D-Bbox	50.9077 45.9942 42.0627	70.2602 44.4837 42.9534	55.0510 37.5991 32.3090	5060ti x1
PointPillars_kitti	BEV	59.1745 54.3432 50.5029	84.4268 67.1347 63.7455	89.9664 87.9104 85.7638	
PointPillars_kitti_snow	BEV	56.2570 50.6632 47.2526	71.4513 46.5849 44.2080	74.9016 50.8983 49.5825	
PointPillars_kitti	2D-BBox	64.6159 61.3713 57.6031	86.2569 73.0707 70.1706	90.6471 89.3323 86.6528	
PointPillars_kitti_snow	2D-BBox	56.9435 51.8297 49.3069	76.5018 52.2800 49.3459	78.9322 59.3901 51.9992	
PointPillars_kitti	AOS	49.2289 46.5926 43.7520	85.0412 69.0913 66.2914	90.4754 88.6834 85.7248	
PointPillars_kitti_snow	AOS	41.4645 38.1470 36.4075	75.5731 50.5951 47.5046	77.9229 58.2583 50.6567	

Repo	Metric	Pedestrian AP@0.5	Cyclist AP@0.5	Car AP@0.7
PointRCNN_kitti	3D-Bbox	70.3937, 62.4997, 56.4812	86.8038, 67.5356, 64.1532	84.0957, 75.6573, 75.8359
PointRCNN_kitti_snow	3D-Bbox	69.2039, 61.6143, 55.8719	79.7640, 56.6199, 53.0094	64.7429, 52.0929, 48.6623
PointRCNN_kitti	BEV	73.7393, 65.3820, 58.8687	87.0454, 69.4767, 65.7828	87.8752, 85.8902, 85.3261
PointRCNN_kitti_snow	BEV	73.4029, 65.4532, 59.0363	83.5402, 58.0971, 54.9253	77.3060, 64.4550, 59.9026
PointRCNN_kitti	2D-BBox	71.4616, 65.1817, 59.4463	89.1654, 72.4245, 68.6297	94.9392, 87.9672, 88.0076
PointRCNN_kitti_snow	2D-BBox	72.7548, 65.3306, 59.3042	83.7134, 58.8943, 56.7163	85.9689, 68.0673, 66.3882
PointRCNN_kitti	AOS	69.10, 62.54, 56.73	89.09, 71.76, 68.01	94.89, 87.81, 87.75
PointRCNN_kitti_snow	AOS	69.94, 62.63, 56.60	83.64, 58.65, 56.48	85.33, 67.30, 65.22

Repo	Metric	Pedestrian AP@0.5	Cyclist AP@0.5	Car AP@0.7
VoxelRCNN_kitti	3D-Bbox	66.5348, 60.2782, 55.6542	85.9466, 72.1112, 68.8626	89.6149, 83.8685, 78.8079
VoxelRCNN_snow	3D-Bbox	66.6909, 59.3986, 55.9992	83.5091, 59.1099, 55.1060	74.8540, 56.2723, 50.7544
VoxelRCNN_kitti	BEV	67.7843, 63.0610, 59.7228	92.3742, 74.5277, 72.2761	90.2165, 88.1085, 87.6400
VoxelRCNN_snow	BEV	70.8167, 64.0834, 58.8447	87.0567, 60.0769, 57.7704	79.2096, 66.7585, 60.1988
VoxelRCNN_kitti	2D-BBox	75.4673, 71.4710, 67.2322	94.7069, 82.7996, 80.4283	97.5992, 89.6159, 89.2497
VoxelRCNN_snow	2D-BBox	75.7812, 68.8897, 66.4404	88.2552, 64.6207, 60.7499	89.7385, 71.0313, 70.0774
VoxelRCNN_kitti	AOS	70.46, 66.24, 62.10	94.51, 82.32, 79.90	97.55, 89.50, 89.06
VoxelRCNN_snow	AOS	70.80, 64.04, 61.58	87.88, 64.09, 60.24	89.37, 70.42, 69.21

Camera-LiDAR fusion

Repo	Metric	Pedestrian AP@0.5	Cyclist AP@0.5	Car AP@0.7
MVMM_kitti	3D-Bbox	58.0598, 54.6385, 51.5646	76.9789, 63.7341, 61.3158	88.7584, 78.1791, 77.2052
MVMM_kitti_snow	3D-Bbox	53.9292, 52.8448, 50.2185	71.5915, 50.4884, 47.6168	74.8244, 54.8129, 50.4703
MVMM_kitti	BEV	61.9839, 58.5879, 55.5531	82.6586, 69.0585, 65.9598	90.1502, 87.6711, 86.3628
MVMM_kitti_snow	BEV	60.4902, 58.1564, 55.3361	78.2400, 55.0631, 52.3427	83.7933, 65.5323, 60.4207
MVMM_kitti	2D-BBox	66.6428, 64.5338, 62.9940	89.8408, 77.7837, 75.5444	94.6267, 89.7352, 89.1435
MVMM_kitti_snow	2D-BBox	66.9772, 64.8389, 62.6820	82.8715, 60.3086, 58.3068	87.8108, 69.2236, 67.2224
MVMM_kitti	AOS	61.9944, 59.8307, 57.8615	89.6691, 76.8684, 74.5149	94.5895, 89.5287, 88.8404
MVMM_kitti_snow	AOS	62.7076, 60.2084, 57.7439	82.6868, 59.9001, 57.8186	87.5130, 68.5216, 66.3796

- LiDAR-only 3DOD 모델의 3d-bbox metric에서 cyclist, car는 성능 하락이 분명하게 보임. 하지만 pedestrian의 경우 성능 하락이 크게 보이지 않음.
반면, Fusion 모델인 MVMM에서는 pedestrian의 경우도 성능이 하락한 것을 확인할 수 있음.
-> 카메라의 노이즈가 더욱 크게 작용했을 거라고 추측 중

해당 원인 분석을 위해 3d-bbox를 이미지에 출력해봄

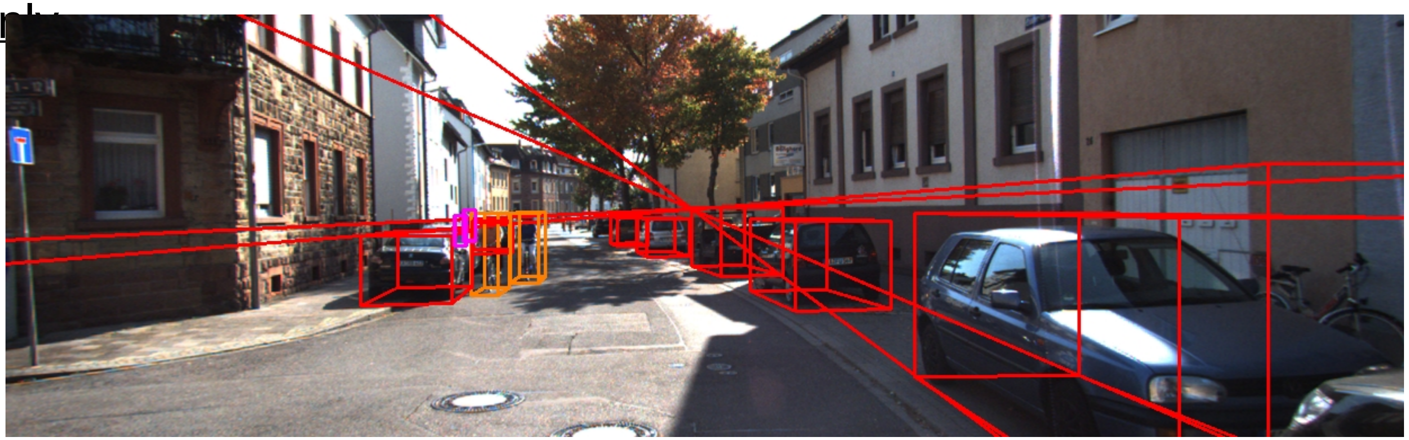
Why doesn't snow simulation affect Pedestrian much?
- LiDAR-only (VoxelRCNN)

Why doesn't snow simulation affect Pedestrian much? - LiDAR-only

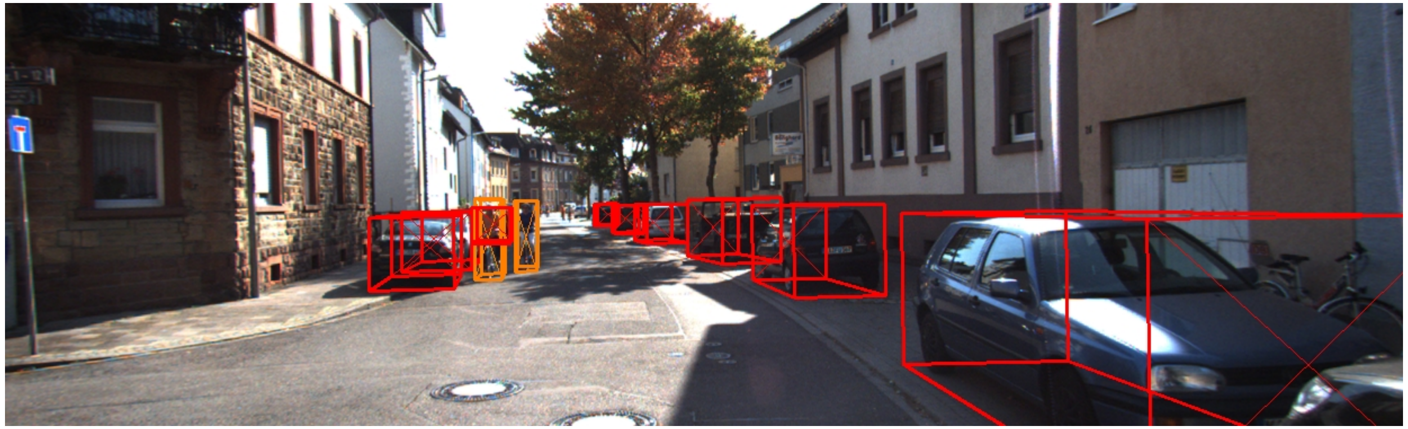
VoxelRCNN : 000201

	Car	Pedestrian	Cyclist
GT	초록색	하늘색(cyan)	노란색
Pred	빨간색	보라색(magenta)	주황색

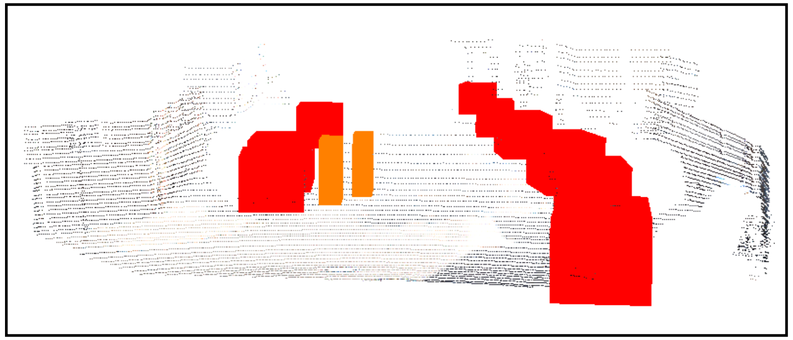
(1) Kitti_GT
(2) Kitti_pred : prediction(3d-bbox) of raw-kitti
(3) Snow_pred : prediction(3d-bbox) of snow-kitti



(1)



(2)



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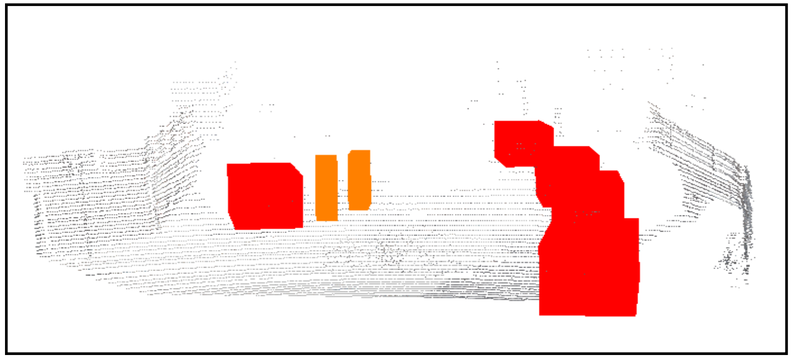
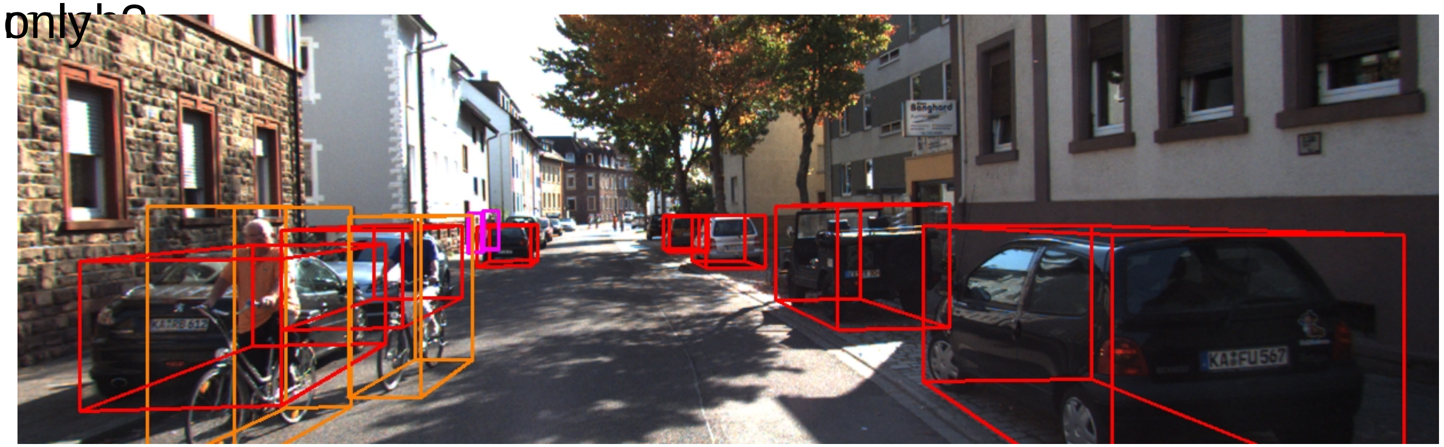


Why doesn't snow simulation affect Pedestrian much? - LiDAR-only'b?

VoxelRCNN : 001221

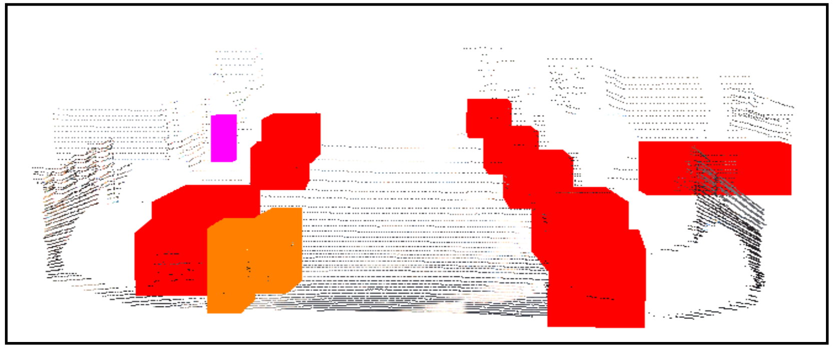
	Car	Pedestrian	Cyclist
GT	초록색	하늘색(cyan)	노란색
Pred	빨간색	보라색(magenta)	주황색

(1) Kitti_GT
(2) Kitti_pred : prediction(3d-bbox) of raw-kitti
(3) Snow_pred : prediction(3d-bbox) of snow-kitti

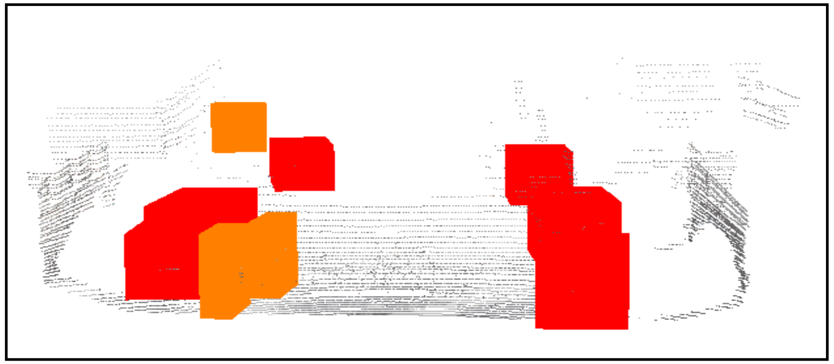


(1)

(2)



(3)



Why doesn't snow simulation affect Pedestrian much? - LiDAR-only?

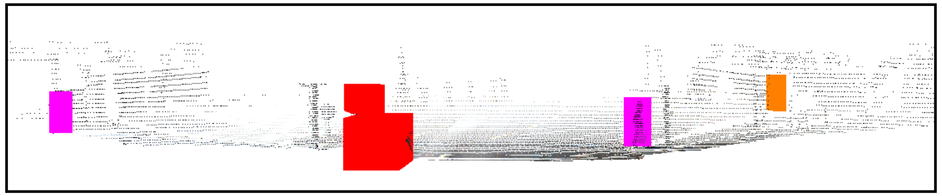
VoxelRCNN : 006024

	Car	Pedestrian	Cyclist
GT	초록색	하늘색(cyan)	노란색
Pred	빨간색	보라색(magenta)	주황색

(1) Kitti_GT
(2) Kitti_pred : prediction(3d-bbox) of raw-kitti
(3) Snow_pred : prediction(3d-bbox) of snow-kitti

(1)

(2)



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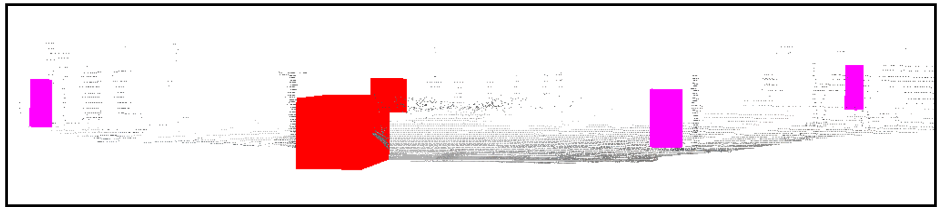
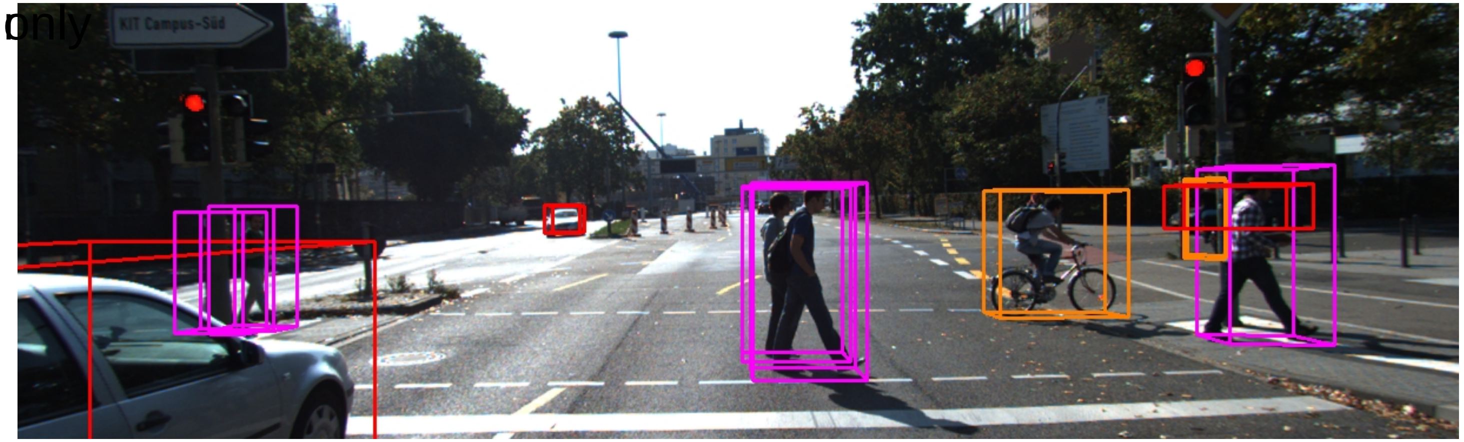


VoxelRCNN : 006496

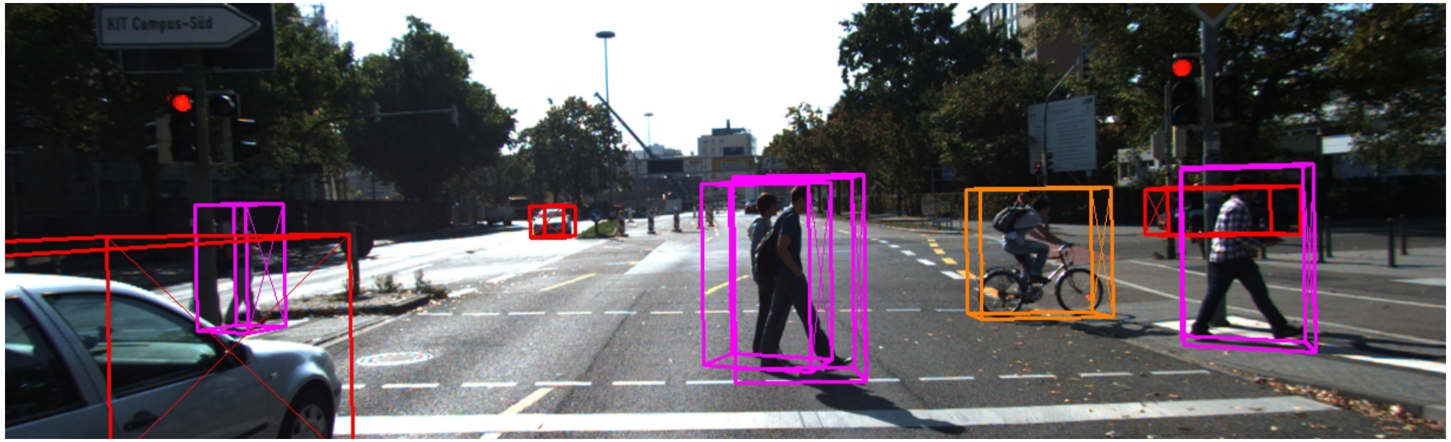
Why doesn't snow simulation affect Pedestrian much? - LiDAR-only

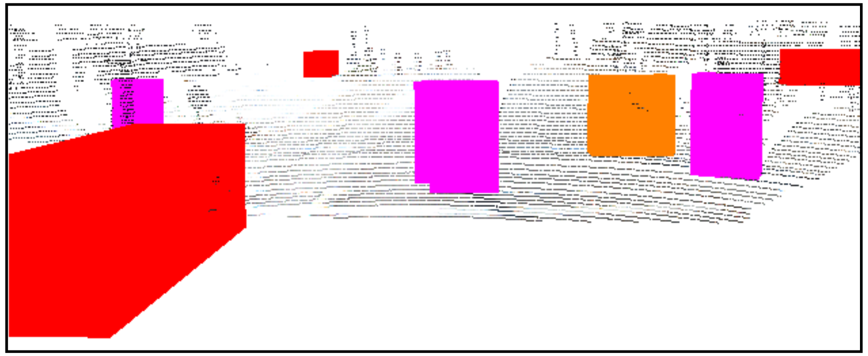
	Car	Pedestrian	Cyclist
GT	초록색	하늘색(cyan)	노란색
Pred	빨간색	보라색(magenta)	주황색

(1) Kitti_GT
(2) Kitti_pred : prediction(3d-bbox) of raw-kitti
(3) Snow_pred : prediction(3d-bbox) of snow-kitti

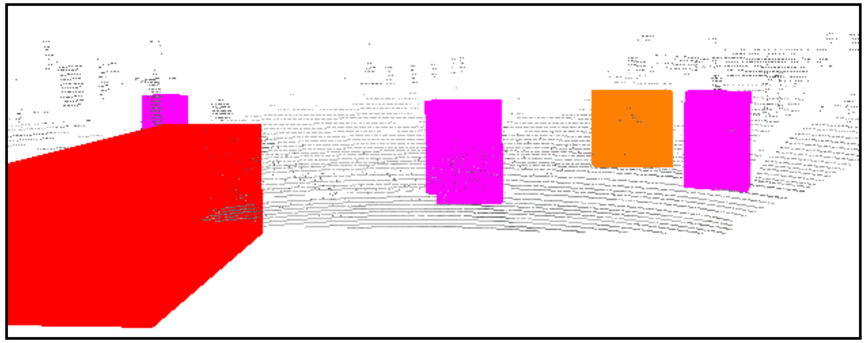


(1)





(2)



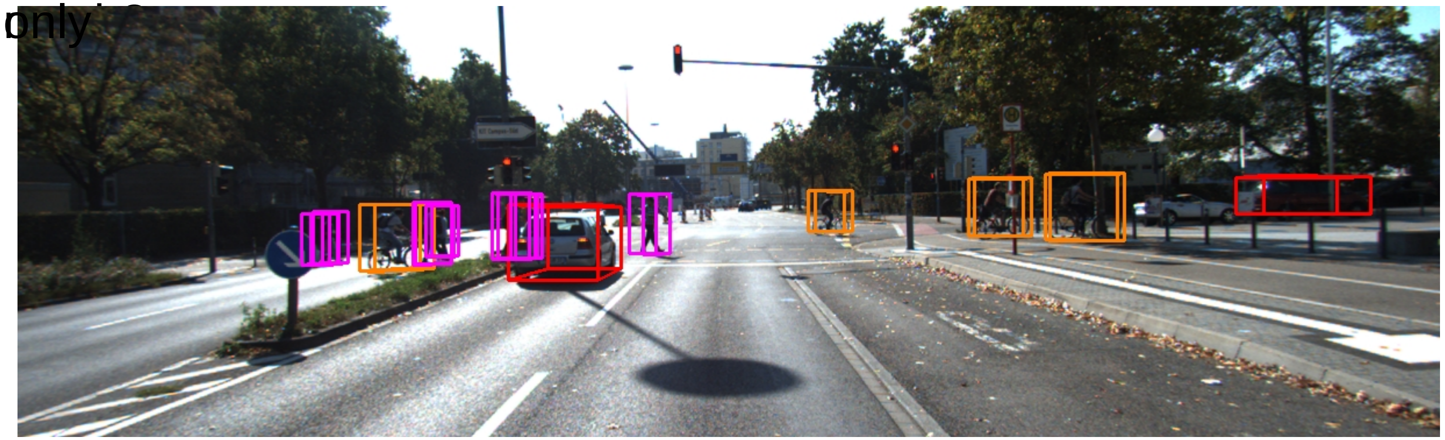
(3)

Why doesn't snow simulation affect Pedestrian much? - LiDAR-only ?

VoxelRCNN : 006813

	Car	Pedestrian	Cyclist
GT	초록색	하늘색(cyan)	노란색
Pred	빨간색	보라색(magenta)	주황색

(1) Kitti_GT
(2) Kitti_pred : prediction(3d-bbox) of raw-kitti
(3) Snow_pred : prediction(3d-bbox) of snow-kitti



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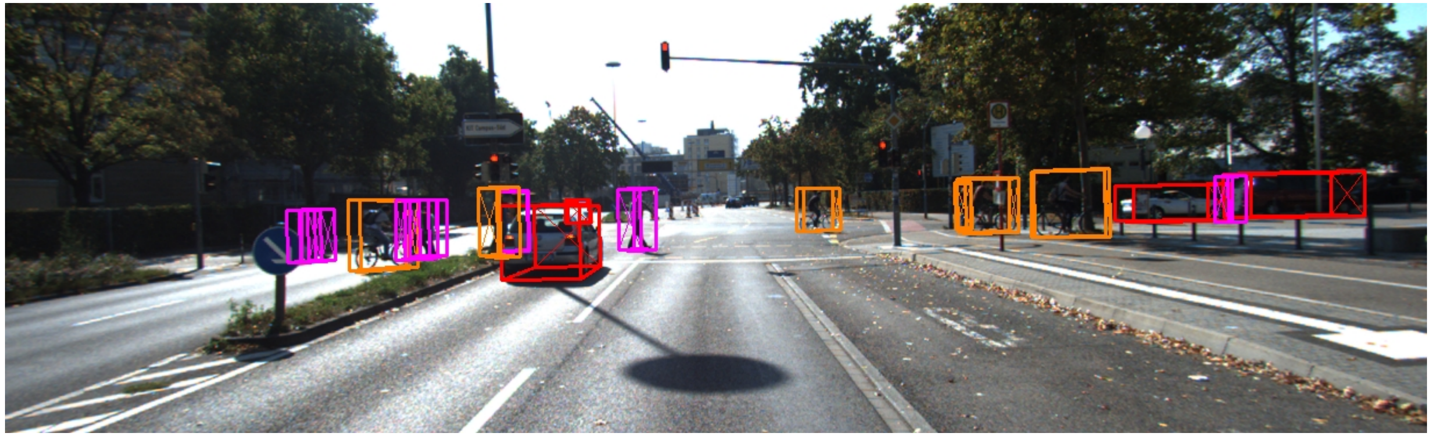
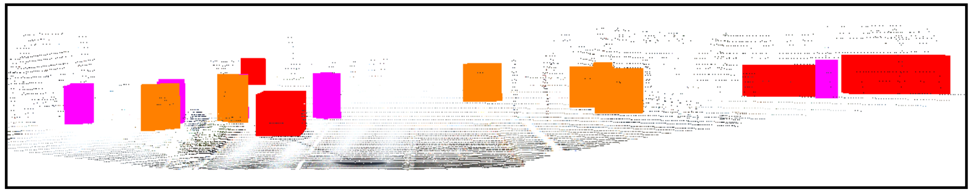
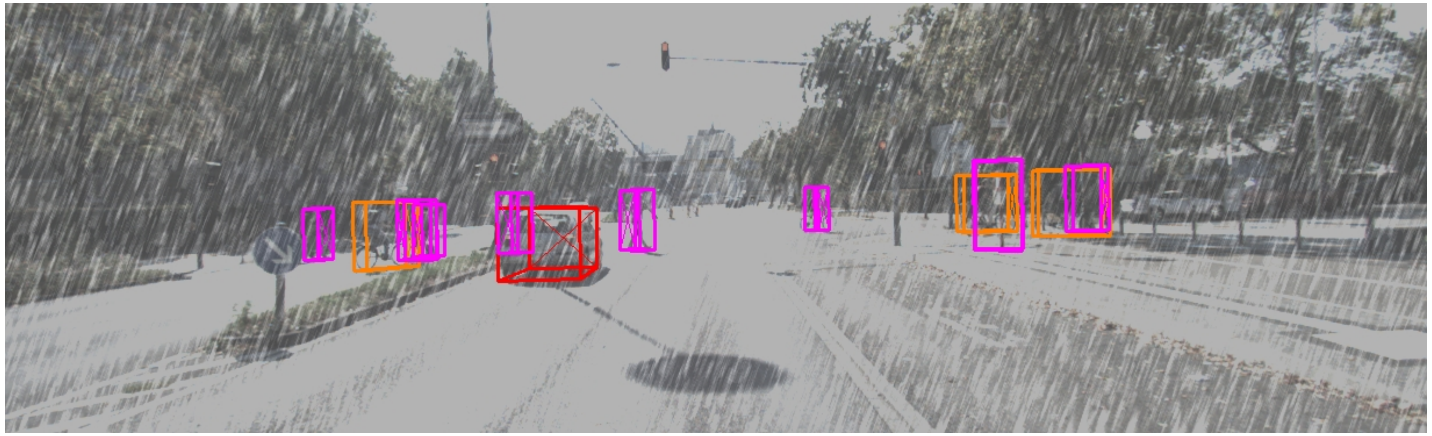
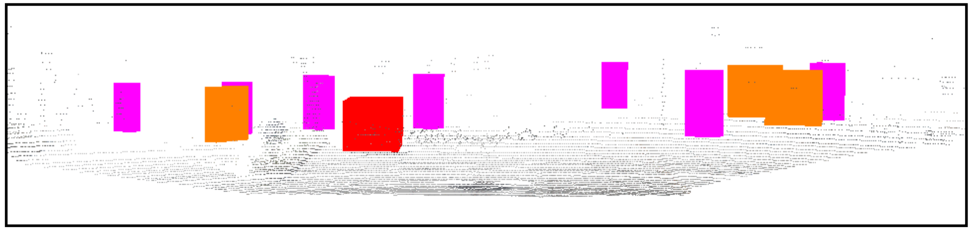




(2)



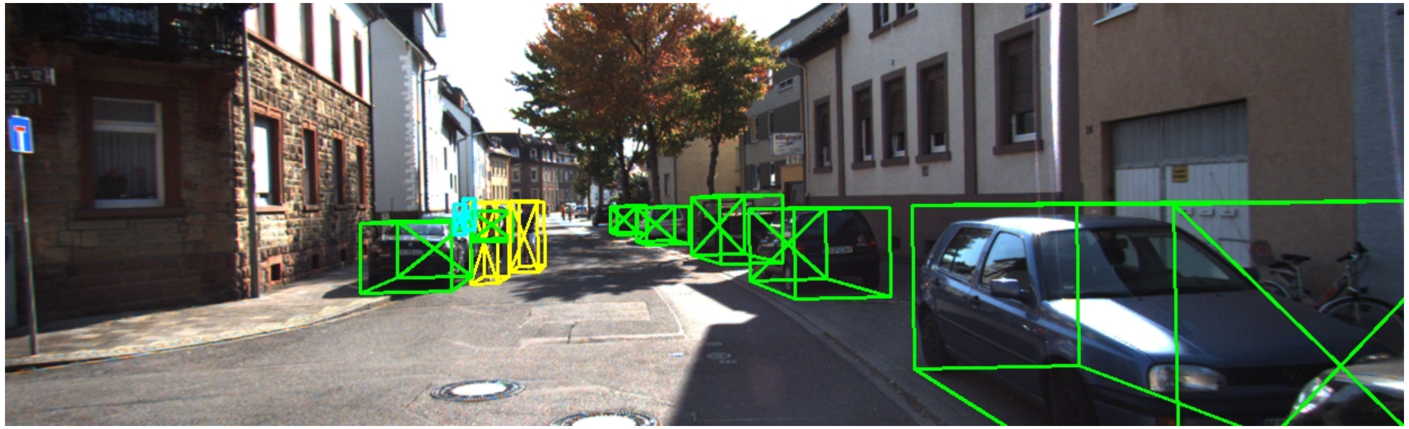


(3)

Why doesn't snow simulation affect Pedestrian much?
- LiDAR-Camera Fusion (MVMM)

Why doesn't snow simulation affect Pedestrian much? - Fusion

MVMM : 000201



(1
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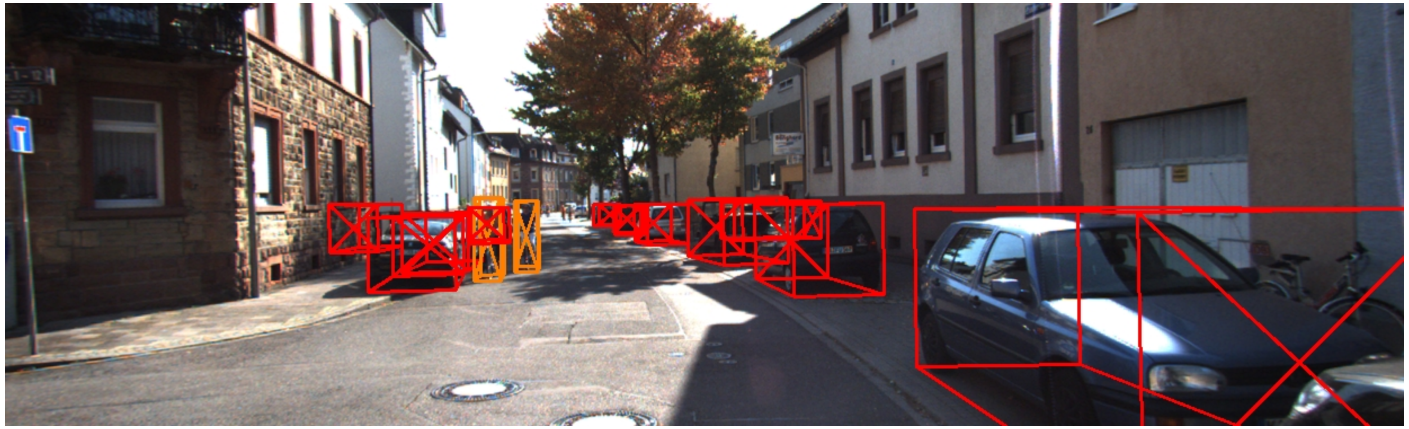

(2
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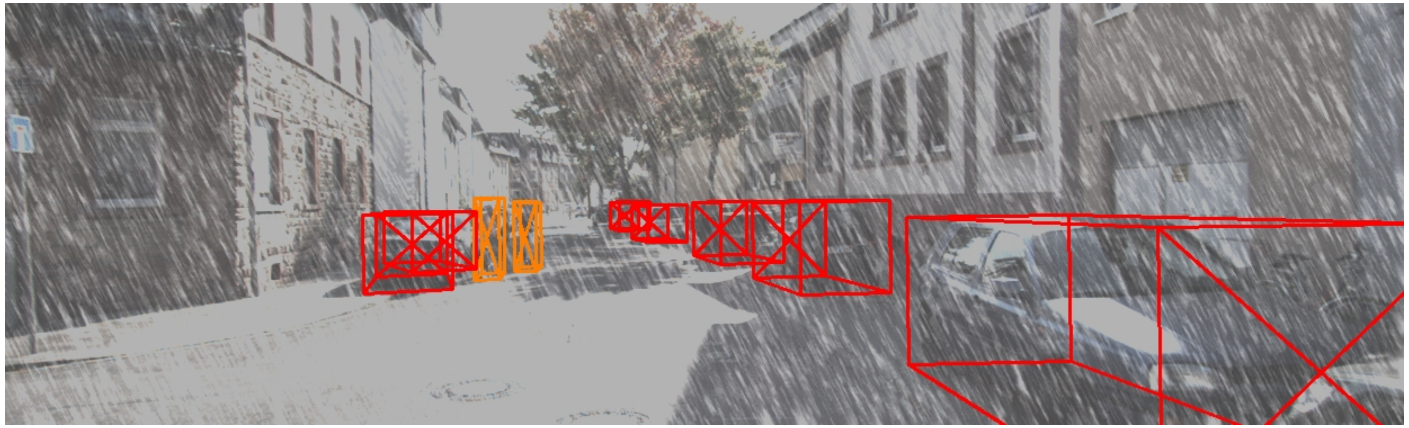
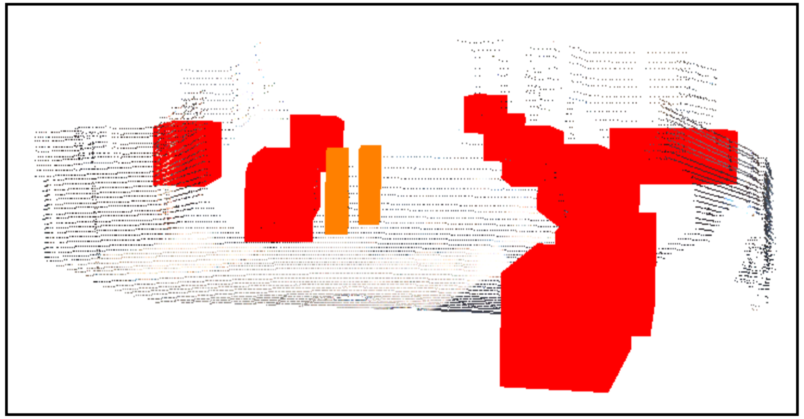
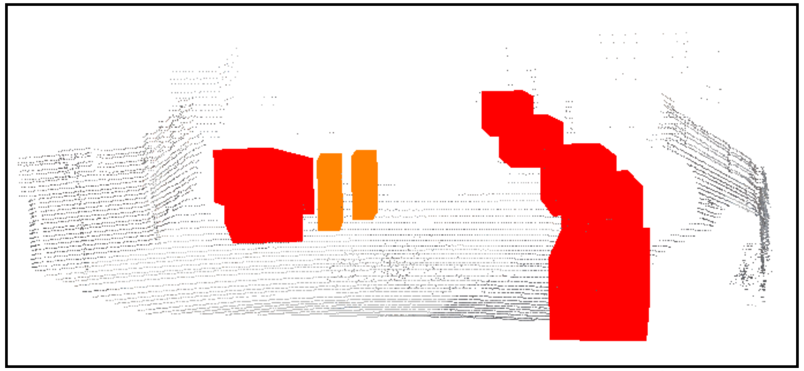


(3
)

	Car	Pedestrian	Cyclist
GT	초록색	하늘색(cyan)	노란색
Pred	빨간색	보라색(magenta)	주황색

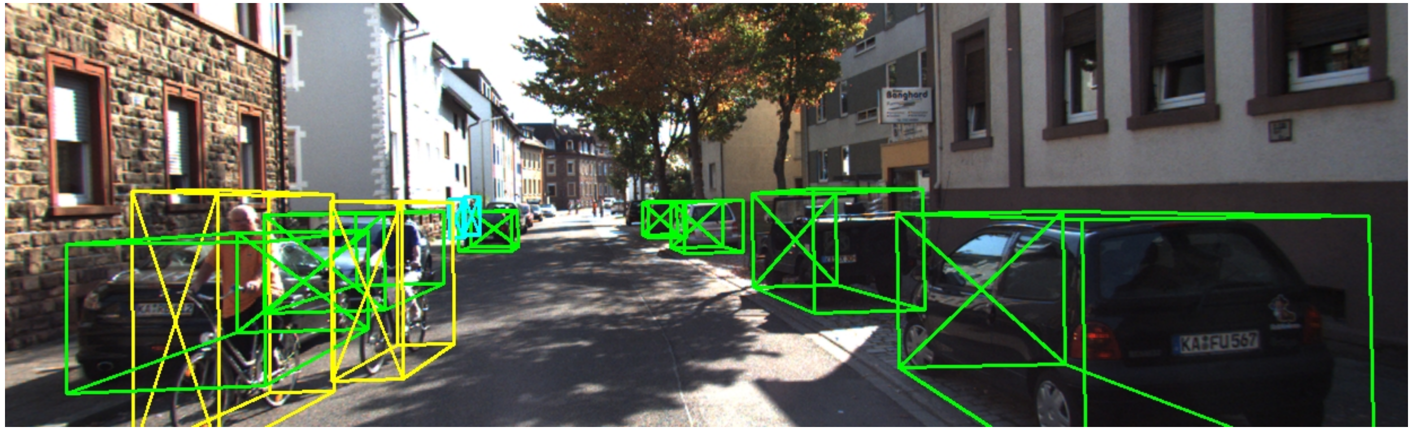
(1) Kitti_GT
(2) Kitti_pred : prediction(3d-bbox) of raw-kitti
(3) Snow_pred : prediction(3d-bbox) of snow-kitti





Why doesn't snow simulation affect Pedestrian much? - Fusion

MVMM : 001221



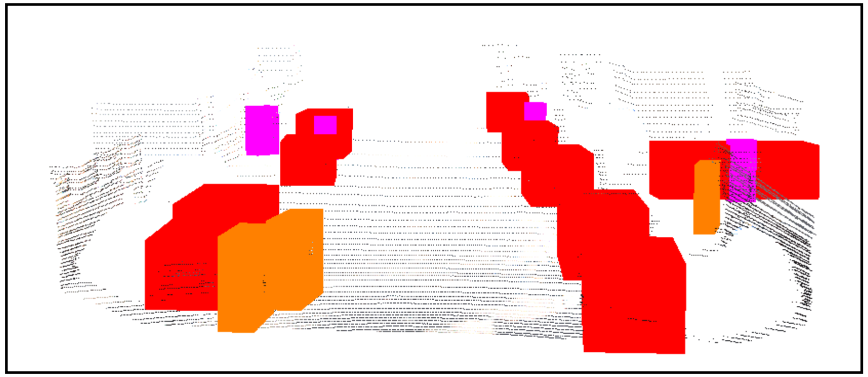
(1)

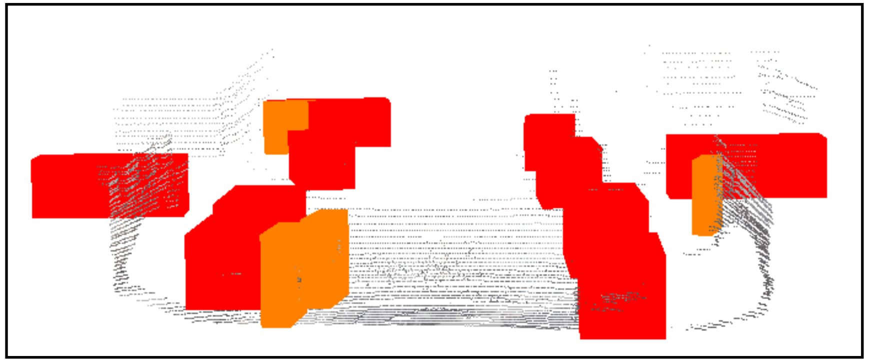
(2)

(3)

	Car	Pedestrian	Cyclist
GT	초록색	하늘색(cyan)	노란색
Pred	빨간색	보라색(magenta)	주황색

(1) Kitti_GT
(2) Kitti_pred : prediction(3d-bbox) of raw-kitti
(3) Snow_pred : prediction(3d-bbox) of snow-kitti





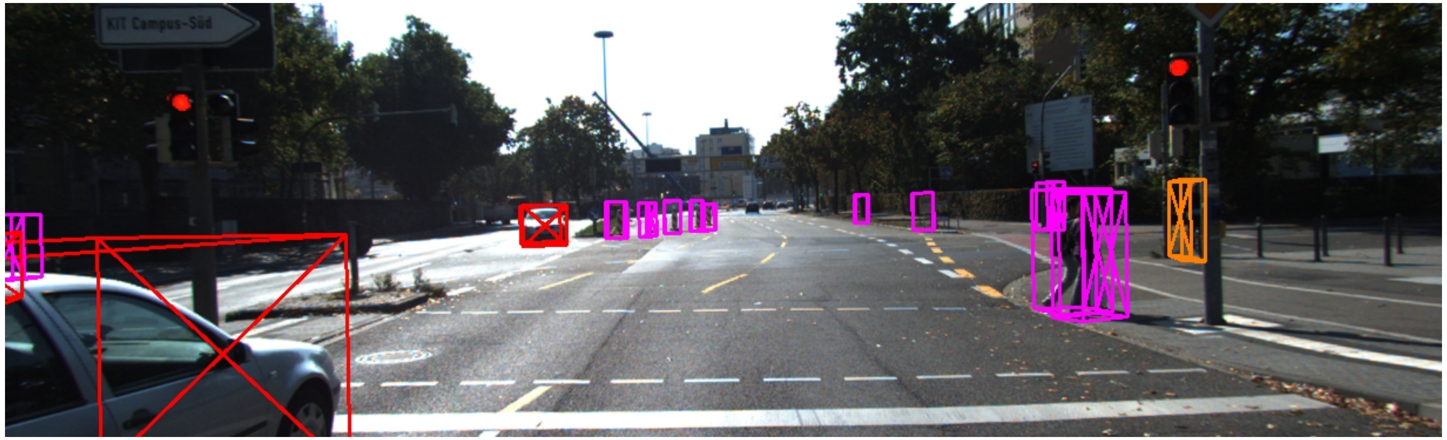
Why doesn't snow simulation affect Pedestrian much? - <u>Fusion</u>

MVMM : 006024

	Car	Pedestrian	Cyclist
GT	초록색	하늘색(cyan)	노란색
Pred	빨간색	보라색(magenta)	주황색

(1) Kitti_GT
(2) Kitti_pred : prediction(3d-bbox) of raw-kitti
(3) Snow_pred : prediction(3d-bbox) of snow-kitti

(1)



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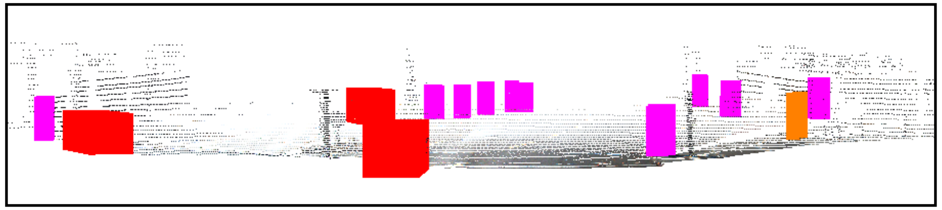
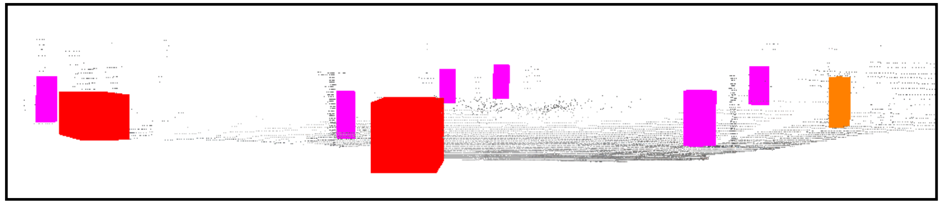
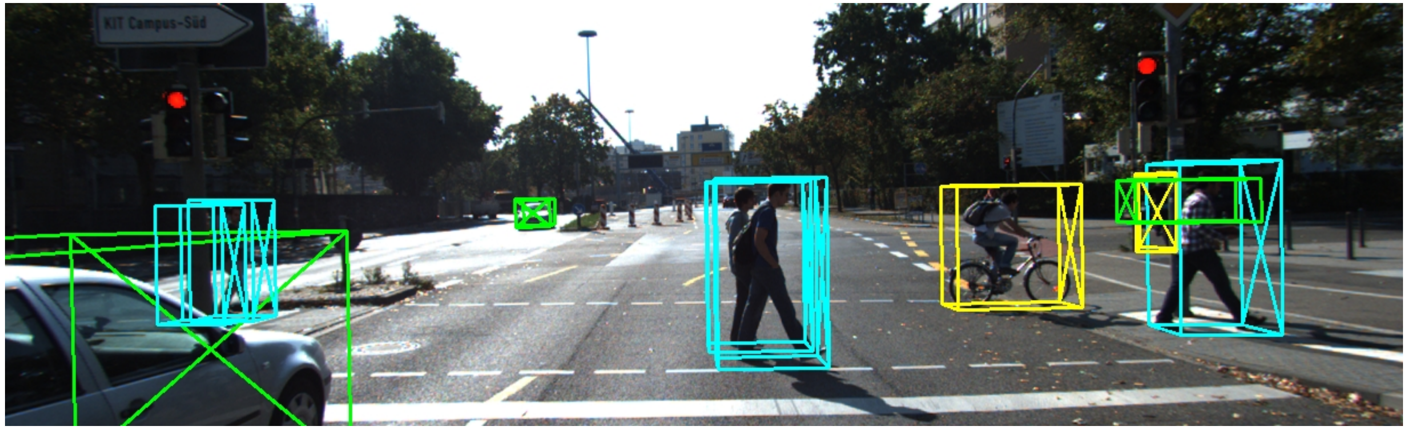


(3)

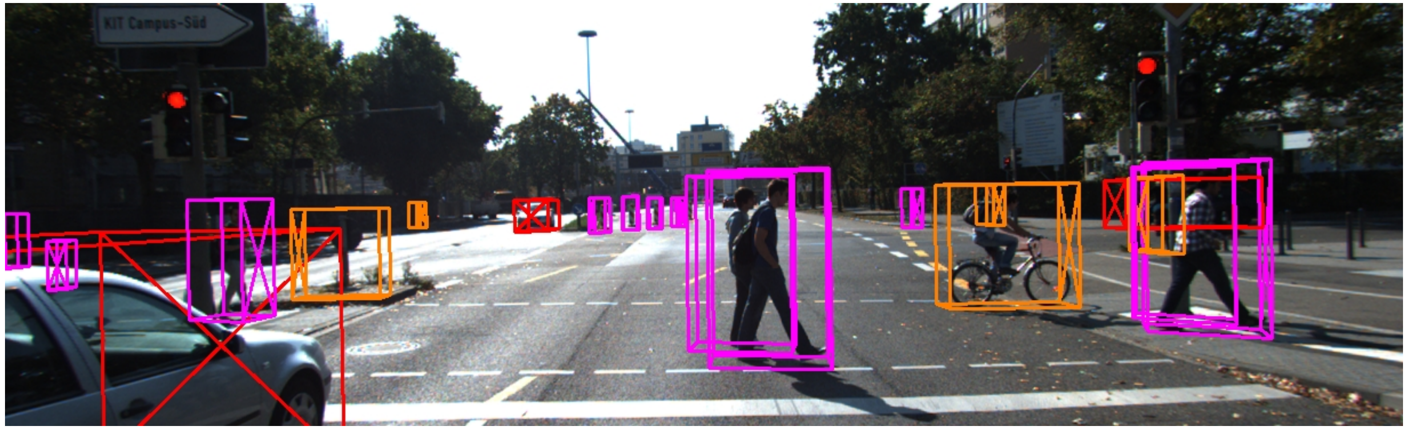


Why doesn't snow simulation affect Pedestrian much? - <u>Fusion</u>

MVMM : 006496



(1)



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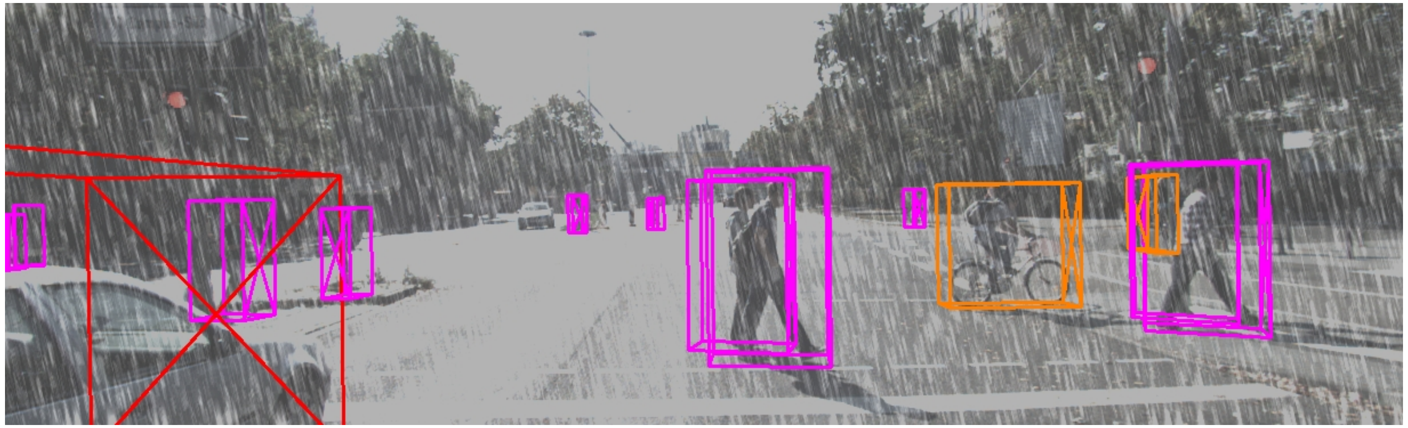
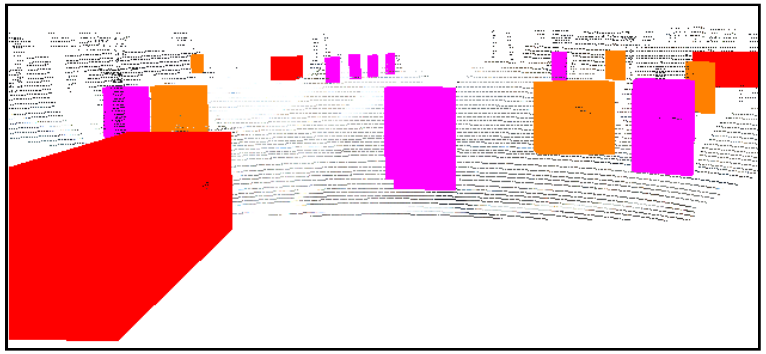
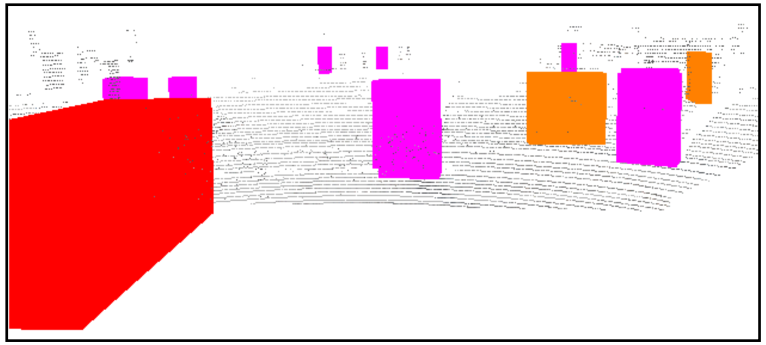


(3)

	Car	Pedestrian	Cyclist
GT	초록색	하늘색(cyan)	노란색
Pred	빨간색	보라색(magenta)	주황색

(1) Kitti_GT
(2) Kitti_pred : prediction(3d-bbox) of raw-kitti
(3) Snow_pred : prediction(3d-bbox) of snow-kitti



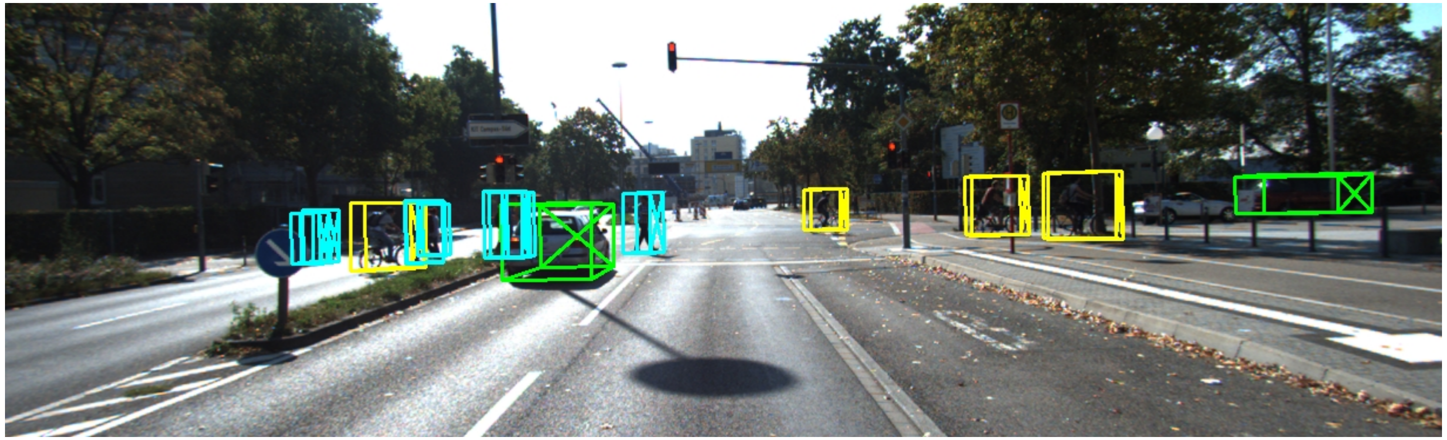


Why doesn't snow simulation affect Pedestrian much? - Fusion

MVMM : 006813

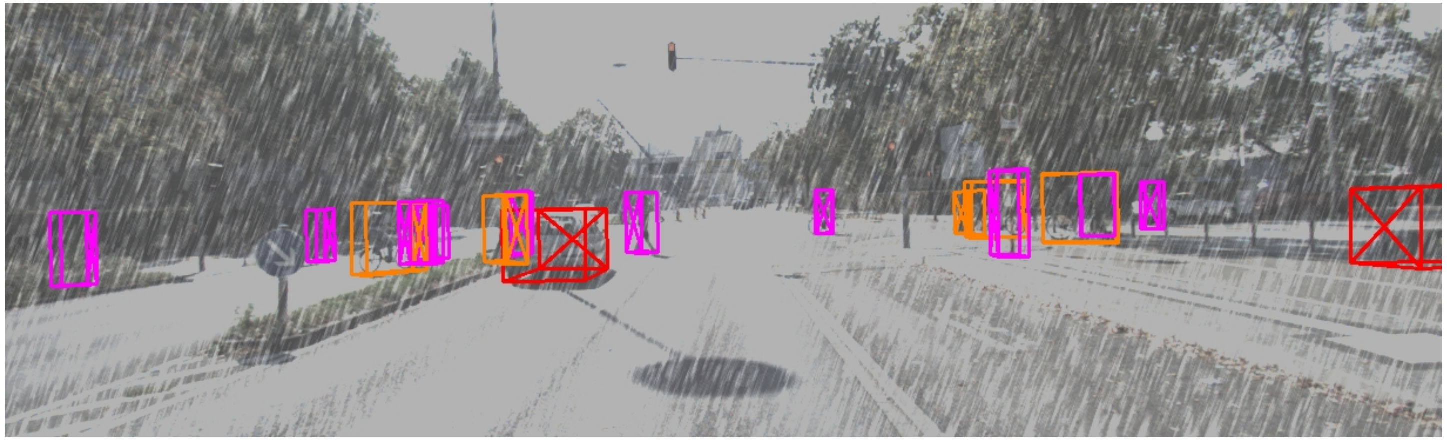
	Car	Pedestrian	Cyclist
GT	초록색	하늘색(cyan)	노란색
Pred	빨간색	보라색(magenta)	주황색

(1) Kitti_GT
(2) Kitti_pred : prediction(3d-bbox) of raw-kitti
(3) Snow_pred : prediction(3d-bbox) of snow-kitti

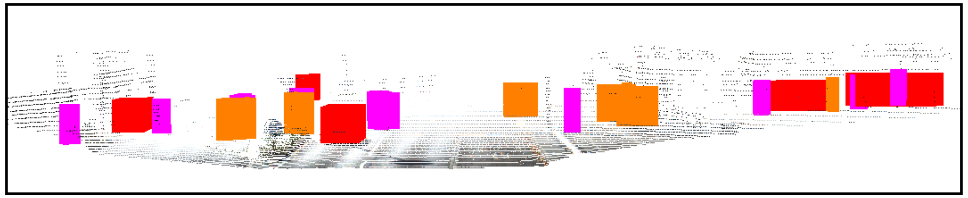


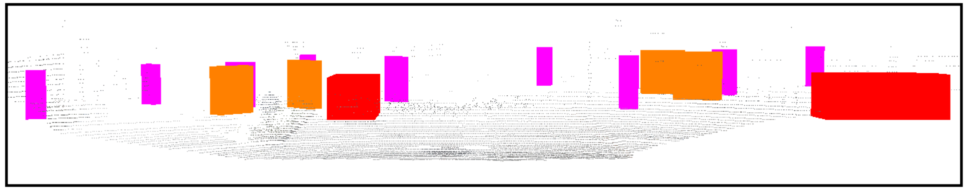
(1)

(2)



(3)





Why doesn't snow simulation affect Pedestrian much?
- Comparison VoxelRCNN vs MVMM

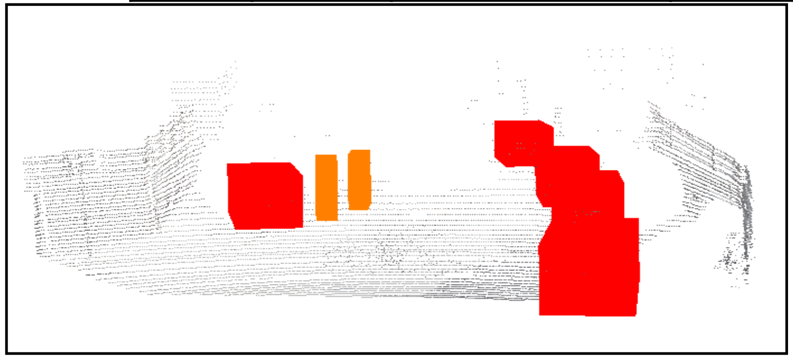
LiDAR-based / fusion

Why doesn't snow simulation affect Pedestrian much?

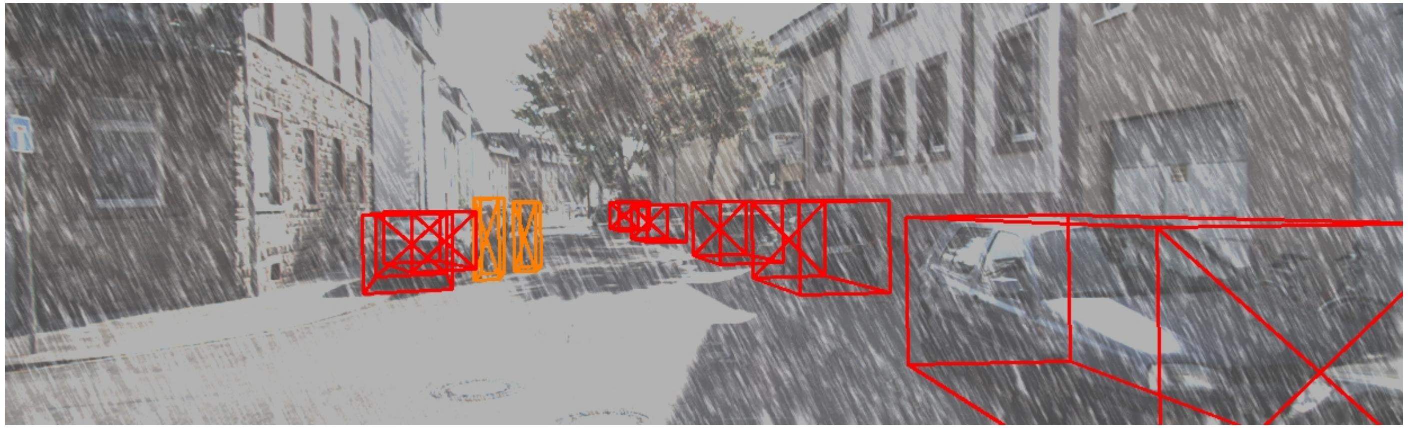
- LiDAR-based(VoxelRCNN) vs Fusion(MVMM)

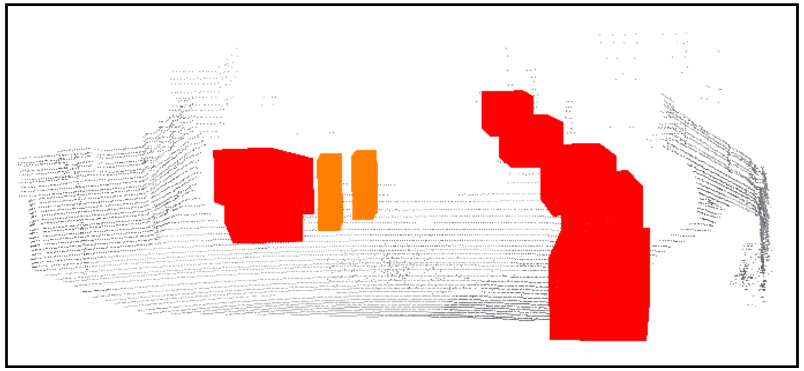
000201

	Car	Pedestrian	Cyclist
GT	초록색	하늘색(cyan)	노란색
Pred	빨간색	보라색(magenta)	주황색



VoxelRCN
N





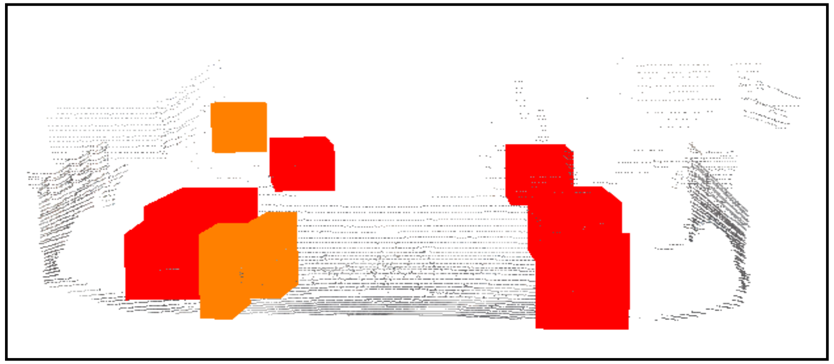
MVMM

Why doesn't snow simulation affect Pedestrian much?

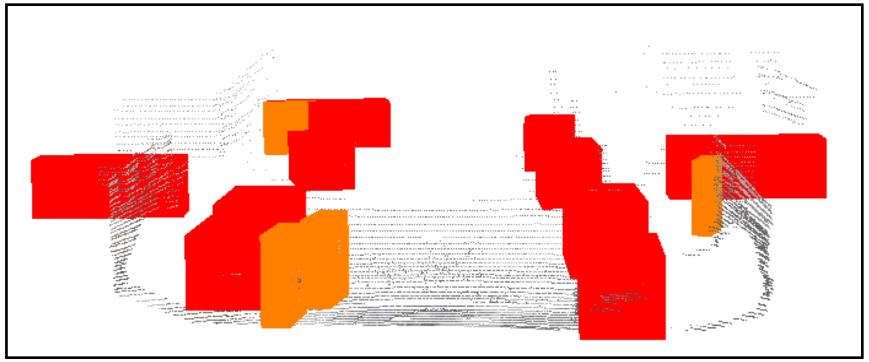
- LiDAR-based(VoxelRCNN) vs Fusion(MVMM)

001221

	Car	Pedestrian	Cyclist
GT	초록색	하늘색(cyan)	노란색
Pred	빨간색	보라색(magenta)	주황색



VoxelRCNN



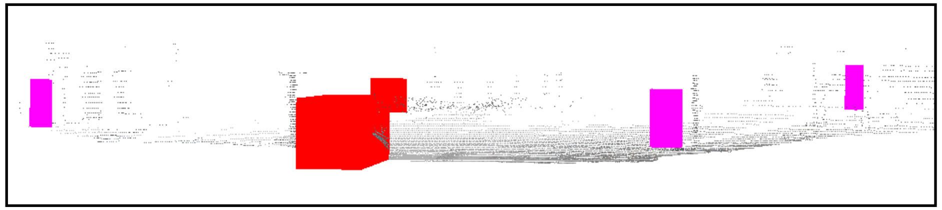
MVMM

Why doesn't snow simulation affect Pedestrian much?

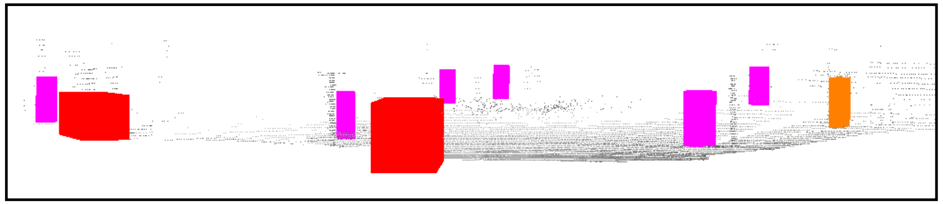
- LiDAR-based(VoxelRCNN) vs Fusion(MVMM)

006024

	Car	Pedestrian	Cyclist
GT	초록색	하늘색(cyan)	노란색
Pred	빨간색	보라색(magenta)	주황색



VoxelRCN
N



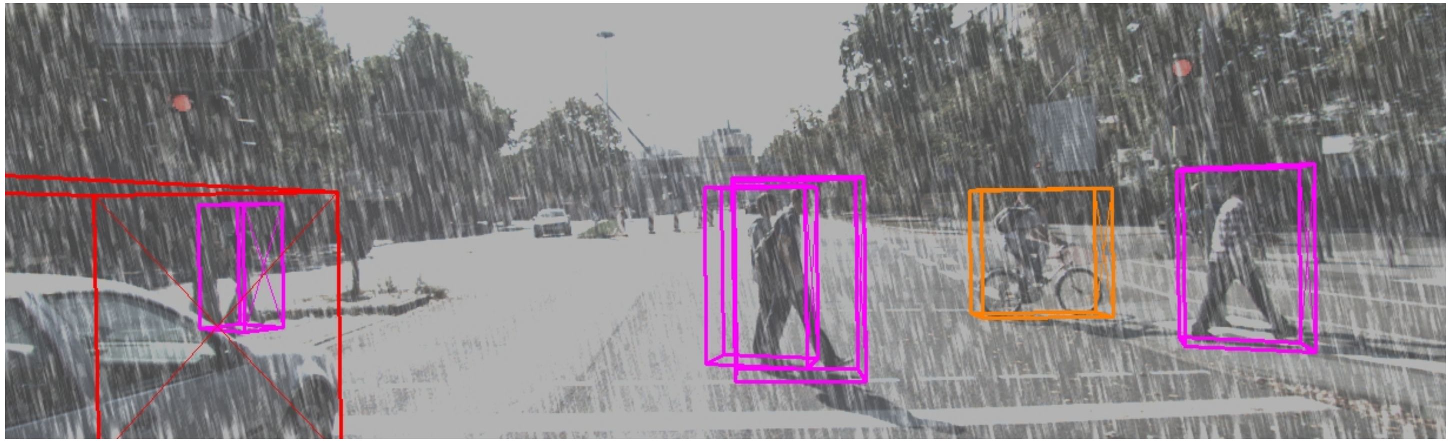
MVMM

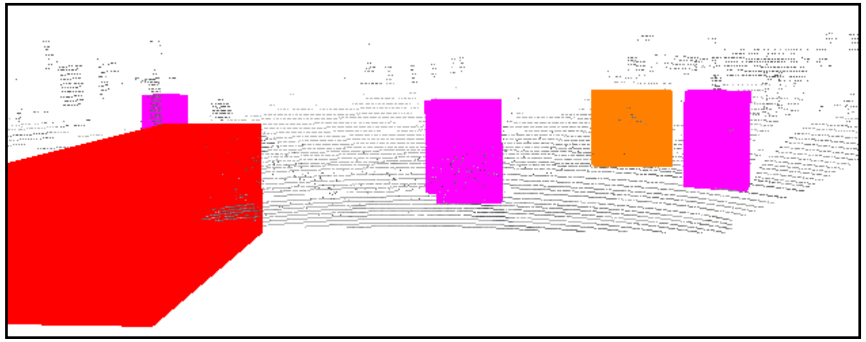
Why doesn't snow simulation affect Pedestrian much?

- LiDAR-based(VoxelRCNN) vs Fusion(MVMM)

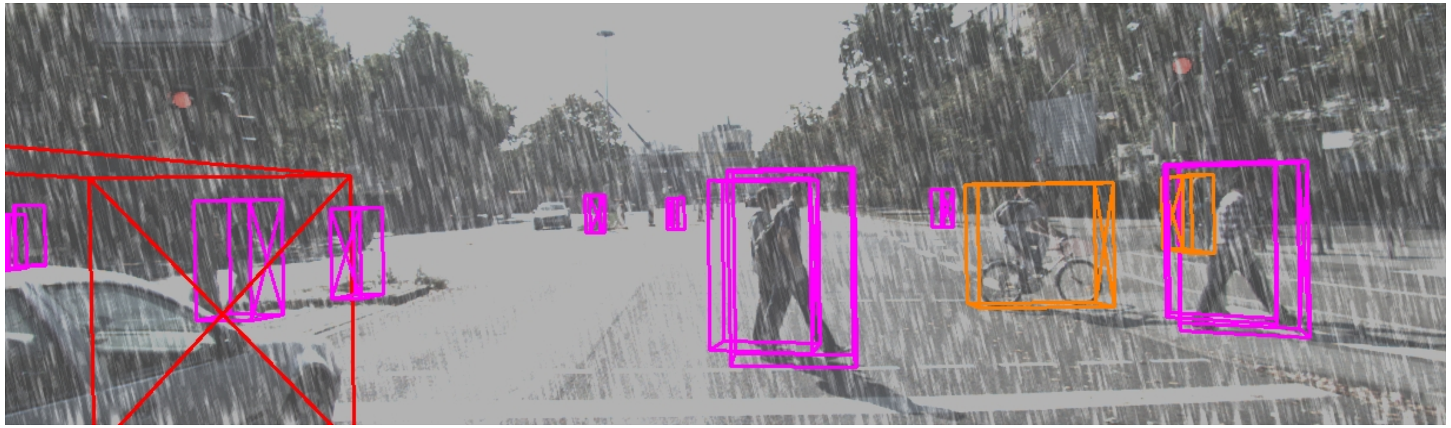
006496

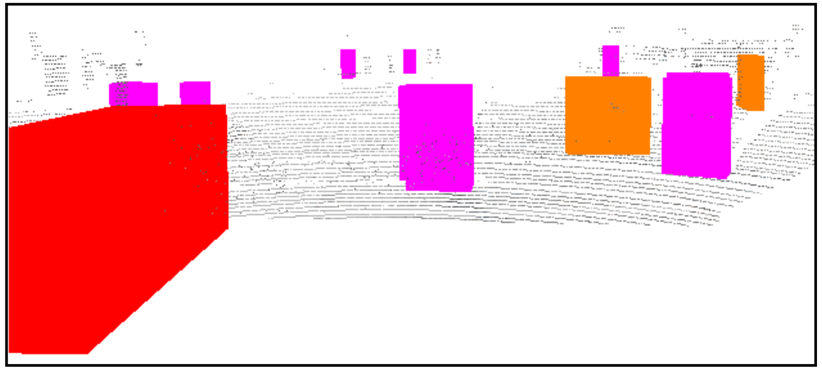
	Car	Pedestrian	Cyclist
GT	초록색	하늘색(cyan)	노란색
Pred	빨간색	보라색(magenta)	주황색





VoxelRCNN





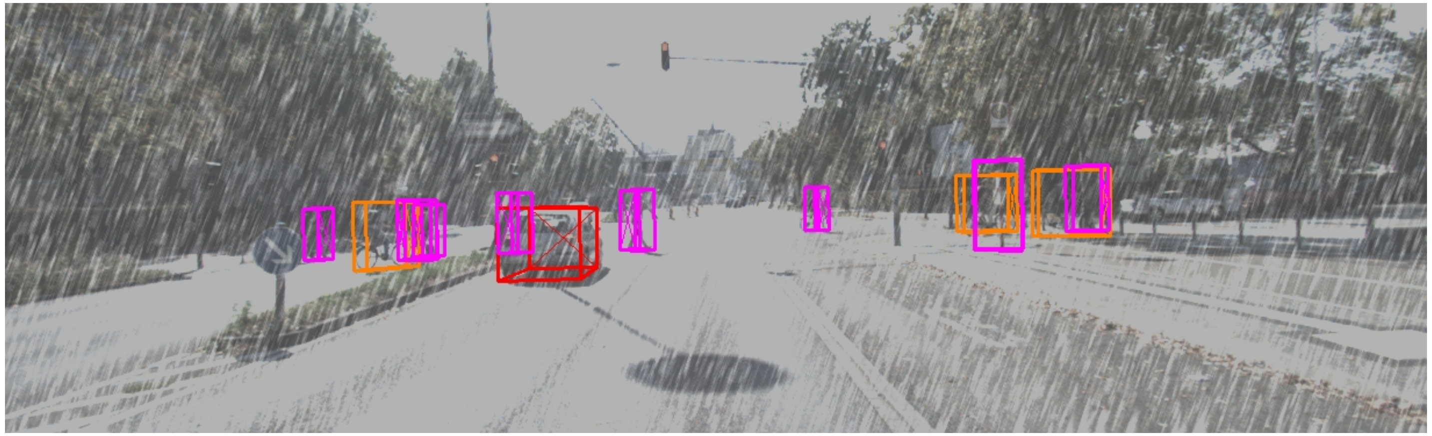
MVMM

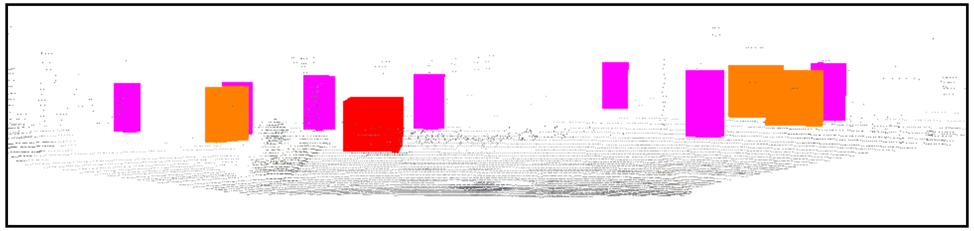
Why doesn't snow simulation affect Pedestrian much?

- LiDAR-based(VoxelRCNN) vs Fusion(MVMM)

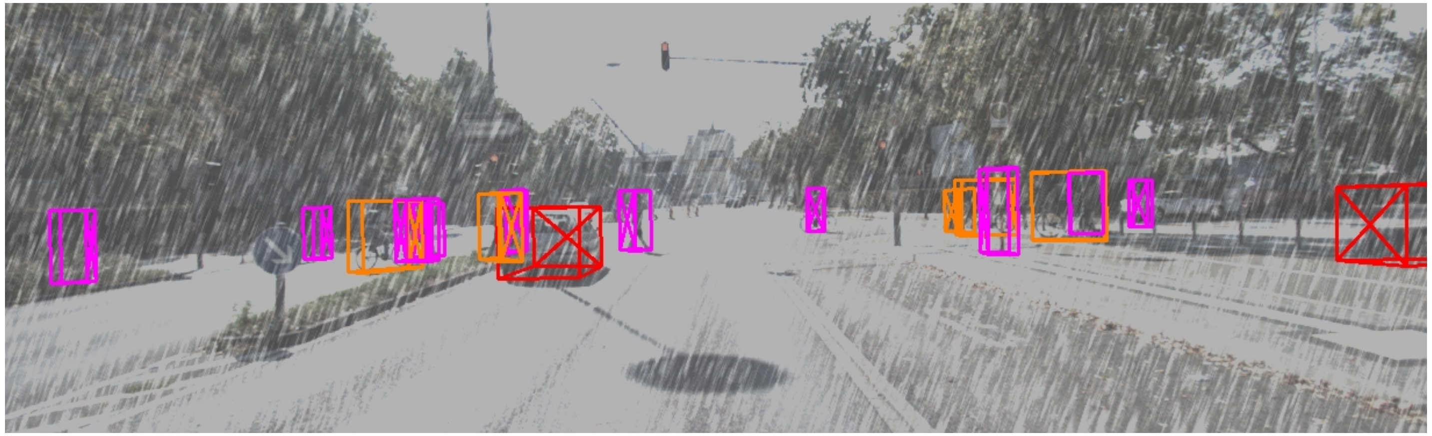
006813

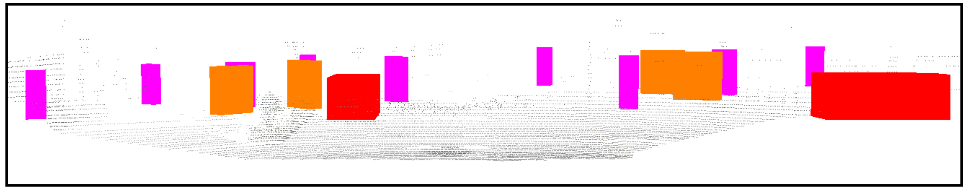
	Car	Pedestrian	Cyclist
GT	초록색	하늘색(cyan)	노란색
Pred	빨간색	보라색(magenta)	주황색





VoxelRCNN





MVMM

Why doesn't snow simulation affect Pedestrian much?
- Point cloud removing percentage

Why doesn't snow simulation affect Pedestrian much?

- Point cloud removing percentage

```
================ [ Class-wise Point Density Analysis ] ================
[Car] (총 14305개 유효 객체)
  - 원본 평균 포인트 수 : 499.93 pts
  - 눈(Snow) 평균 포인트 수: 240.46 pts
  - 평균 포인트 유실률  : 63.04 %
-----------------------------------------------------------------------
[Pedestrian] (총 2279개 유효 객체)
  - 원본 평균 포인트 수 : 209.89 pts
  - 눈(Snow) 평균 포인트 수: 193.26 pts
  - 평균 포인트 유실률  : 11.85 %
-----------------------------------------------------------------------
[Cyclist] (총 892개 유효 객체)
  - 원본 평균 포인트 수 : 164.58 pts
  - 눈(Snow) 평균 포인트 수: 135.17 pts
  - 평균 포인트 유실률  : 36.53 %
-----------------------------------------------------------------------
```

- Raw와 snow 처리 후의 point cloud에 대해 GT bbox 범위 내의 유실률을 체크해봤습니다.
- 파란색 박스의 결과는 전체 유실률을 확인한 것이고, 오른쪽은 시각화했던 scene들에 대해 개별 point 유실률을 체크한 것입니다.
- 포인트 유실률이 평균 11.75%로 가장 낮다보니, prediction 에도 큰 영향을 미치지 않은 것으로 해석할 수 있을 것이라 생각했습니다.

```
================ [ Scene 000201 개별 객체 분석 ] ================
Obj 01 [Car        ] : 원본 2724개 -> 눈 적용  538개 | 유실률:  80.2%
Obj 02 [Car        ] : 원본 2315개 -> 눈 적용  850개 | 유실률:  63.3%
Obj 03 [Car        ] : 원본  639개 -> 눈 적용  307개 | 유실률:  52.0%
Obj 04 [Cyclist    ] : 원본  122개 -> 눈 적용  108개 | 유실률:  11.5%
Obj 05 [Car        ] : 원본  305개 -> 눈 적용  104개 | 유실률:  65.9%
Obj 06 [Car        ] : 원본  247개 -> 눈 적용  135개 | 유실률:  45.3%
Obj 07 [Car        ] : 원본  112개 -> 눈 적용   62개 | 유실률:  44.6%
Obj 08 [Car        ] : 원본    4개 -> 눈 적용    1개 | 유실률:  75.0%
Obj 09 [Car        ] : 원본   41개 -> 눈 적용    8개 | 유실률:  80.5%
Obj 10 [Pedestrian] : 원본   10개 -> 눈 적용    3개 | 유실률:  70.0%
Obj 11 [Pedestrian] : 원본    9개 -> 눈 적용    2개 | 유실률:  77.8%
Obj 12 [Cyclist    ] : 원본   81개 -> 눈 적용   67개 | 유실률:  17.3%
=================================================================
```

```
================ [ Scene 006024 개별 객체 분석 ] ================
Obj 01 [Car        ] : 원본 6323개 -> 눈 적용 3288개 | 유실률:  48.0%
Obj 02 [Cyclist    ] : 원본   35개 -> 눈 적용   32개 | 유실률:   8.6%
Obj 03 [Pedestrian] : 원본  107개 -> 눈 적용  103개 | 유실률:   3.7%
Obj 04 [Pedestrian] : 원본  188개 -> 눈 적용  187개 | 유실률:   0.5%
Obj 05 [Car        ] : 원본   59개 -> 눈 적용   10개 | 유실률:  83.1%
=================================================================
```

```
================ [ Scene 006496 개별 객체 분석 ] ================
Obj 01 [Car        ] : 원본 6271개 -> 눈 적용 3266개 | 유실률:  47.9%
Obj 02 [Pedestrian] : 원본  308개 -> 눈 적용  296개 | 유실률:   3.9%
Obj 03 [Pedestrian] : 원본  138개 -> 눈 적용  118개 | 유실률:  14.5%
Obj 04 [Pedestrian] : 원본  458개 -> 눈 적용  437개 | 유실률:   4.6%
Obj 05 [Cyclist    ] : 원본   36개 -> 눈 적용   31개 | 유실률:  13.9%
Obj 06 [Pedestrian] : 원본  153개 -> 눈 적용  148개 | 유실률:   3.3%
Obj 07 [Pedestrian] : 원본   35개 -> 눈 적용   29개 | 유실률:  17.1%
Obj 08 [Cyclist    ] : 원본  338개 -> 눈 적용  320개 | 유실률:   5.3%
Obj 09 [Car        ] : 원본   34개 -> 눈 적용    1개 | 유실률:  97.1%
Obj 10 [Car        ] : 원본   28개 -> 눈 적용    1개 | 유실률:  96.4%
=================================================================
```

```
================ [ Scene 001221 개별 객체 분석 ] ================
Obj 01 [Car        ] : 원본 1841개 -> 눈 적용  965개 | 유실률:  47.6%
Obj 02 [Cyclist    ] : 원본  638개 -> 눈 적용  558개 | 유실률:  12.5%
Obj 03 [Car        ] : 원본  476개 -> 눈 적용  197개 | 유실률:  58.6%
Obj 04 [Car        ] : 원본 1018개 -> 눈 적용  739개 | 유실률:  27.4%
Obj 05 [Car        ] : 원본  274개 -> 눈 적용  102개 | 유실률:  62.8%
Obj 06 [Car        ] : 원본  207개 -> 눈 적용  130개 | 유실률:  37.2%
Obj 07 [Car        ] : 원본   82개 -> 눈 적용   26개 | 유실률:  68.3%
Obj 08 [Car        ] : 원본   56개 -> 눈 적용   28개 | 유실률:  50.0%
Obj 09 [Pedestrian] : 원본   12개 -> 눈 적용    8개 | 유실률:  33.3%
Obj 10 [Pedestrian] : 원본   21개 -> 눈 적용   11개 | 유실률:  47.6%
Obj 11 [Cyclist    ] : 원본  475개 -> 눈 적용  407개 | 유실률:  14.3%
=================================================================
```

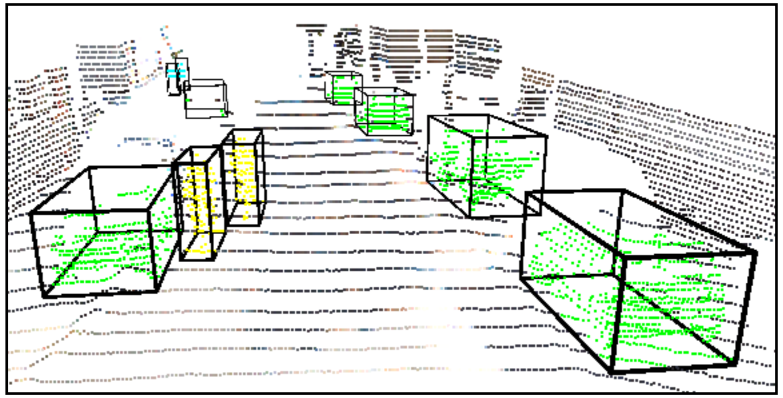
```
================ [ Scene 006813 개별 객체 분석 ] ================
Obj 01 [Car        ] : 원본  296개 -> 눈 적용  167개 | 유실률:  43.6%
Obj 02 [Cyclist    ] : 원본  104개 -> 눈 적용   85개 | 유실률:  18.3%
Obj 03 [Cyclist    ] : 원본   60개 -> 눈 적용   45개 | 유실률:  25.0%
Obj 04 [Pedestrian] : 원본   59개 -> 눈 적용   45개 | 유실률:  23.7%
Obj 05 [Cyclist    ] : 원본   40개 -> 눈 적용   12개 | 유실률:  70.0%
Obj 06 [Pedestrian] : 원본   51개 -> 눈 적용   40개 | 유실률:  21.6%
Obj 07 [Pedestrian] : 원본   26개 -> 눈 적용   22개 | 유실률:  15.4%
Obj 08 [Pedestrian] : 원본   17개 -> 눈 적용   14개 | 유실률:  17.6%
Obj 09 [Cyclist    ] : 원본  107개 -> 눈 적용   75개 | 유실률:  29.9%
Obj 10 [Pedestrian] : 원본   20개 -> 눈 적용   18개 | 유실률:  10.0%
Obj 11 [Pedestrian] : 원본   56개 -> 눈 적용   43개 | 유실률:  23.2%
Obj 12 [Pedestrian] : 원본   26개 -> 눈 적용   21개 | 유실률:  19.2%
Obj 13 [Car        ] : 원본   64개 -> 눈 적용    8개 | 유실률:  87.5%
=================================================================
```

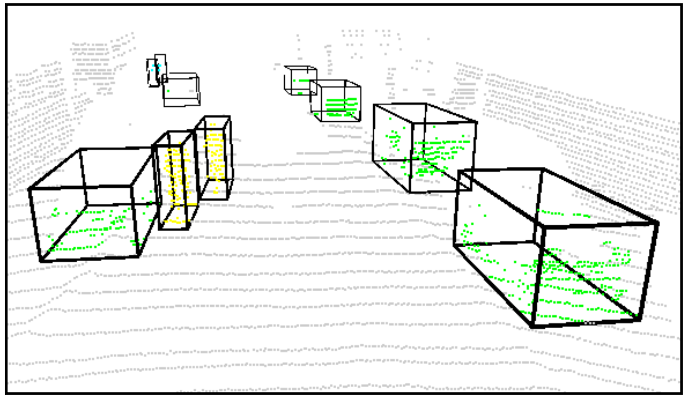
Why doesn't snow simulation affect Pedestrian much?

Scene - 000201

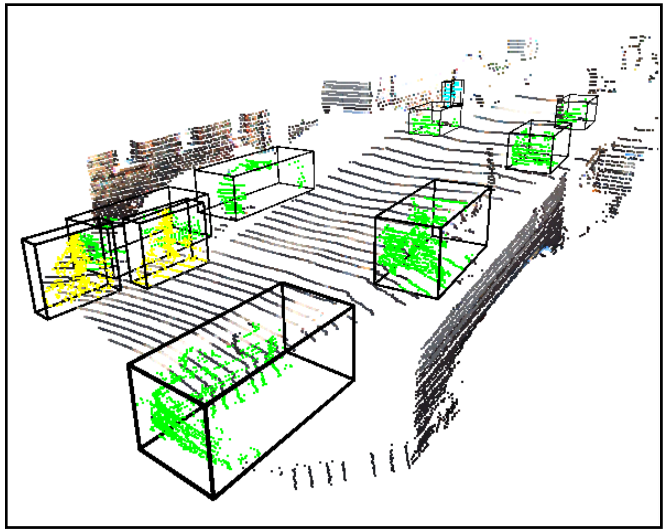
Raw kitti point cloud

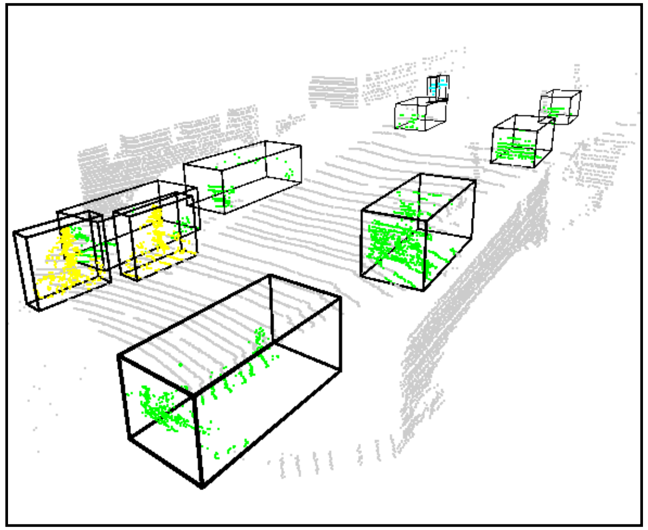
Snow kitti point cloud





Scene - 001221



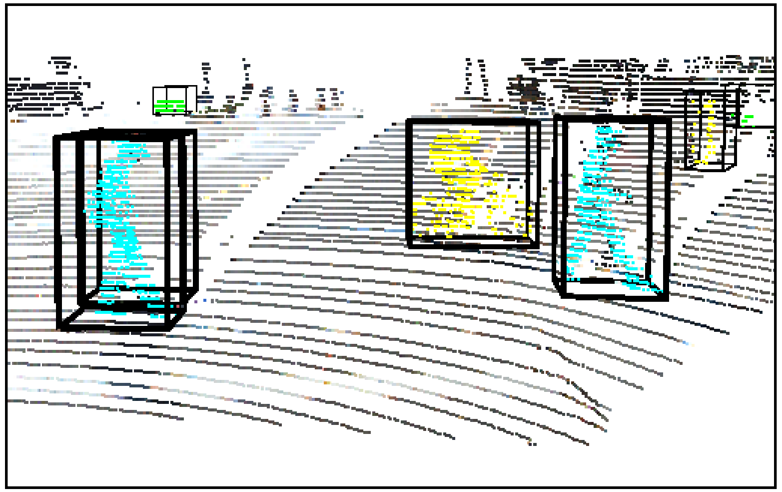


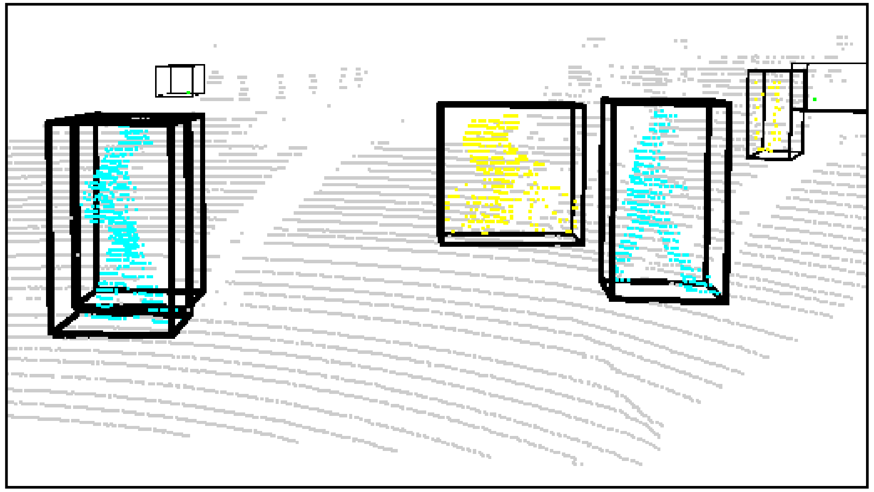
Why doesn't snow simulation affect Pedestrian much?

Scene - 006496

Raw kitti point cloud

Snow kitti point cloud





Scene - 006813